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Optical Facial Fit

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COMSATS University Islamabad, Sahiwal Campus

Optical Facial Fit

**A project presented to
COMSATS Institute of Information Technology, Islamabad**

**In partial fulfillment
Of the requirement for the degree of
Bachelor of Science in Computer Science (2021-2025)**

By

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DECLARATION

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CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS (CS) “**Optical Facial Fit**” was developed by **Muhammad Talal Saeed (CIIT/FA21-BCS-059/SWL)** **Syed Muhammad Ali (CIIT/FA21-BCS-064/SWL)** under the supervision of “**Dr.Fakhar Mustafa**” and co- supervisor “**Ms.Fatima Khan**” and that in their opinion, it is fully adequate, in scope and quality for the degree of Bachelor of Science in Computer Sciences.

Supervisor

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Executive Summary

Optical Facial Fit is a full-featured web platform that we have personally developed to redefine the eyewear shopping experience by blending AI technology with e-commerce. During our research and observation of user behavior in the optical industry, We discovered that many people struggle with choosing the right frames online due to the inability to try them on virtually. Optical Facial Fit is my solution to bridge this gap by offering an AI-powered virtual try-on feature, enabling users to see how different glasses look on their faces in real-time before making a purchase.

The platform integrates several core modules: **Facial Recognition & AI Try-On, Product Catalog, User Registration, Cart and Order Placement**. Users can seamlessly register, browse a curated collection of stylish frames, use the AI-enhanced try-on feature to see themselves wearing each pair, and then proceed to place an order all within a smooth, interactive interface. The AI try-on functionality uses facial mapping to ensure accurate and realistic visualization, providing users with confidence in their choices.

Throughout the development of Optical Facial Fit, We followed the **Software Development Life Cycle (SDLC), Incremental Model**, starting with the essential features and then expanding the platform with advanced modules like personalized suggestions, user reviews, and order tracking. This phased approach allowed me to continuously test, refine, and improve the platform based on user feedback and functionality.

This project represents not only my technical capabilities but also our commitment to enhancing user experience and solving real-world problems with technology. Every module, every feature, and every interaction in Optical Facial Fit was built with precision and passion to offer users a modern, convenient, and innovative way to shop for glasses online.

Acknowledgement

All praise is to Almighty Allah who bestowed upon us a minute portion of His boundless knowledge by which we were able to accomplish this challenging task.

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Syed Muhammad Ali

Muhammad Talal Saeed

Abbreviations

Abbreviation	Meaning
OFF	Optical Facial Fit
MVC	Model-View-Controller
HTML	Hypertext Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
PHP	Hypertext Preprocessor
AI	Artificial Intelligence
API	Application Programming Interface
DBMS	Database Management System
HCI	Human-Computer Interaction
ORM	Object-Relational Mapping
SQL	Structured Query Language
JSON	JavaScript Object Notation
CRUD	Create, Read, Update, Delete
REST	Representational State Transfer
UI	User Interface
UX	User Experience
MySQL	My Structured Query Language
UI/UX	User Interface/User Experience
SaaS	Software as a Service
AJAX	Asynchronous JavaScript and XML

IDE	Integrated Development Environment
SRS	Software Require Specification
UT	Unit Testing
FT	Functional Testing
IT	Integration Testing

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Chapter 1

INTRODUCTION

1 Introduction

Optical Facial Fit is a comprehensive web platform I developed to address a common challenge in the online eyewear shopping experience helping users confidently choose the right frames without the ability to try them on physically. As e-commerce continues to grow, particularly in the fashion and optical industries, there is a rising demand for digital tools that replicate the in-store experience. Most existing optical websites provide static images or generic model previews, which fail to offer personalized visualization. Optical Facial Fit bridges this gap by integrating AI-powered virtual try-on technology, allowing users to see how different eyeglass frames look on their own faces in real-time [1].

This platform is built to serve a wide range of users, from fashion-conscious individuals to those seeking functional eyewear solutions. It promotes confident and informed purchasing by enabling users to explore a curated product catalog, register for personalized services, add items to their cart or favorites, and experience real-time frame try-ons using facial recognition and mapping technology.

The goal of this project is to enhance user confidence and satisfaction when buying eyewear online by merging artificial intelligence with e-commerce functionality. More than just a technical solution, Optical Facial Fit represents a user-centric approach to solving real-world shopping pain points through innovation and thoughtful design.

1.1 Brief

Optical Facial Fit provides real-time AI-based virtual try-on, facial recognition, a product catalog, a shopping cart, order placement, and an intuitive admin panel, all structured to enhance user confidence and streamline decision-making. Facial Fit combines practical utility with modern technology to deliver a seamless, intelligent, and engaging shopping journey. The system is divided into the following core modules:

1.1.1 User Registration and Profile Management

This module allows users to easily register and manage their accounts. New users can sign up using their email or social media accounts, while returning users can log in securely to access personalized features such as favorites, previous orders, and saved try-ons. Once registered, users can maintain a profile, manage their details, and track their engagement with the platform.

Admins can view and manage user accounts, including activity logs, queries submitted, and interactions within the system. This helps maintain platform integrity and offers users a secure, responsive experience. . Once registered, users can maintain a profile, manage their details, and track their engagement with the platform.

1.1.2 AI-Powered Virtual Try-On and Facial Recognition

At the heart of Optical Facial Fit is the AI-based try-on feature. Using facial recognition and real-time facial mapping, users can virtually try on different eyeglass frames through their device cameras. The system detects facial landmarks and overlays frames with precision, allowing users to see an accurate visual of how each pair will look on their face.

This module increases customer confidence and reduces return rates by allowing users to "try before they buy" virtually replicating the in-store experience in a digital format. The system detects facial landmarks and overlays frames with precision, allowing users to see an accurate visual of how each pair will look on their face.

1.1.3 Product Catalog and Smart Filtering

This module displays a curated collection of eyeglass frames, complete with images, descriptions, pricing, and specifications. Users can browse through different categories such as gender, brand, shape, size, and color. Smart filters help narrow down the selection based on face shape, trending styles, or user preferences.

Admins can add, update, or delete product listings from the backend panel. The catalog also supports promotional tags like "Best Seller," "New Arrival," or "Recommended," which are algorithmically or manually assigned to highlight relevant products. Smart filters help narrow down the selection based on face shape, trending styles, or user preferences.

1.1.4 Cart, Favorites, and Order Placement

This module allows users to manage their eyewear selections efficiently. Users can add products to their **Favorites** for future reference or directly to their Cart for checkout. The platform supports a simplified and secure order placement system where users can review their items, apply discounts (if any), choose delivery options, and confirm payments.

The order process is designed to be user-friendly and transparent, with email confirmations and real-time status updates. Admins can manage orders, update statuses (e.g., pending, dispatched, delivered), and access user order history from the backend.

1.1.5 Contact and Query Management

This module handles user communication with the platform for product inquiries, support, or feedback. Users can fill out a Contact Us form that automatically logs their messages in the admin panel.

Admins can view, respond to, and archive queries, ensuring timely and effective support. This helps build trust and encourages users to engage confidently with the platform.

1.1.6 Admin Panel and Analytics

The admin dashboard serves as the control center for the platform. It provides full visibility and management capabilities over users, products, orders, logs, and user queries. Admins can track platform performance, generate reports, monitor virtual try-on usage, and view analytics related to traffic, sales, and user behavior.

This module empowers platform administrators to make data-driven decisions, optimize product listings, and enhance the overall shopping experience.

1.1.7 FAQs

To provide immediate assistance, users can access a dynamic FAQs section covering common queries about the virtual try-on, order process, shipping, returns, and more. In future iterations, this can be enhanced with an AI Assistant that answers questions in real-time using natural language processing, improving support efficiency and user engagement.

1.2 Relevance to Course Modules

Optical Facial Fit demonstrates the practical application of several core computer science modules covered throughout my Bachelor of Computer Science degree. It reflects how theoretical knowledge can be transformed into a functional AI-based web platform.

1.2.1 Web Development

The front end of Optical Facial Fit was built using HTML, CSS, JavaScript, React.js, and Bootstrap. These technologies, learned during web development courses, enabled me to design a responsive, intuitive, and cross-platform user interface that supports real-time try-ons and smooth navigation through eyewear products.

1.2.2 Software Engineering

Principles from software engineering such as modular design, incremental development, and requirement analysis were fundamental in structuring the system. These helped me plan, organize, and manage the development phases of multiple modules including try-on, admin panel, and product management.

1.2.3 Database Management Systems

Using MySQL with Laravel's Eloquent ORM, we designed the backend to handle user profiles, product catalogs, try-on logs, favorites, shopping carts, and transaction data. This knowledge came directly from DBMS coursework and optimized relational data models.

1.2.4 Artificial Intelligence (AI)

The AI-driven virtual try-on system is the core of Optical Facial Fit. By applying AI and computer vision principles learned through coursework, we enabled the system to align eyewear over user facial features accurately, providing a realistic and engaging user experience.

1.2.5 Human-Computer Interaction (HCI)

From interface layout to navigation flow, the design was shaped by HCI principles to ensure users could interact comfortably with the platform. Features such as facial try-on previews, tooltips, and simplified product filtering all reflect usability techniques emphasized in HCI courses.

1.3 Project Background

The concept for Optical Facial Fit arose from observing how eyewear shopping still lacks interactivity and personalization, especially in online settings. Customers often hesitate to buy glasses online due to uncertainty about how they will look or fit. This challenge sparked the idea to develop a platform that bridges this gap using AI.

Optical Facial Fit is a web-based platform that enables users to virtually try on eyewear using their device's camera or uploaded image. The platform includes a catalog of products, personalized recommendations, a real-time AI try-on module, shopping cart, and admin-controlled content management. Users can interact with the system in a way that mimics real-life trials, thus increasing confidence in purchasing.

The inclusion of AI, admin moderation, real-time try-on, and social features makes Optical Facial Fit a comprehensive and practical solution for modern e-commerce in the eyewear sector. It serves as a direct response to the real challenges faced by online shoppers.

1.4 Literature Review

Table 1.1 summarizes the gaps in existing online eyewear platforms, such as Lenskart, Warby Parker, highlighting issues such as limited AI integration for virtual try-on, poor user-personalization, fragmented admin control, and insufficient customer interaction management [2]. Optical Facial Fit addresses these gaps by offering an all-in-one solution with advanced AI-based virtual try-on, a comprehensive admin panel, seamless product and contact management, integrated cart and favorites, secure payment methods, and enhanced user engagement features to improve the eyewear shopping experience.

Table 1-1 literature view

Platform	Weakness	Proposed Solution in Optical Facial Fit
Lenskart	Limited customization; AI try-on lacks face mapping precision	Advanced AI alignment for accurate facial fitting of eyewear

Warby Parker	Requires app installation; limited on-browser try-on capabilities	Fully browser-based try-on, no app download required
Amazon Eyewear	Minimal personalization, no AI-assisted fit or sizing recommendations	AI-driven fitting suggestions based on facial geometry and preferences
Standard E-commerce	Product photos only; no interactivity	Real-time AI try-on with webcam or uploaded photo
Online Optical Sites	Lack of admin control over facial try-on logs or interaction tracking	Admin panel for log management, product control, and usage analytics

1.5 Analysis from Literature Review

The analysis highlights gaps in popular eyewear platforms like Lenskart and Warby Parker. These include limited AI precision, lack of real-time browser-based try-on, and no integrated admin control system. Optical Facial Fit solves these issues by offering:

- Advanced AI-based virtual try-on
- Fully browser-accessible interaction (no app needed)
- A structured admin backend for log tracking, user activity, and product control
- Realistic fit and frame adjustment based on user facial dimensions

1.5.1 User Registration and Profile Management

While most e-commerce sites offer registration, few support detailed user profiles for try-on logs, favorites, and personalized preferences. Optical Facial Fit enables users to register and create profiles, save favorites, view try-on history, and manage purchases. Admins can control user data, review logs, and moderate interactions.

1.5.2 AI Virtual Try-On

Existing platforms often lack accurate facial mapping. Our solution uses facial landmark detection and frame fitting algorithms to deliver highly accurate, real-time virtual try-ons. This improves both confidence and engagement during the purchase process.

1.5.3 Product and Cart Management

Unlike generic platforms, Optical Facial Fit includes a modular system for admins to add, update, and categorize eyewear. Users can add items to a cart or favorites list, view details, and proceed to checkout.

1.5.4 Contact and Feedback Module

The contact module allows users to send feedback or queries directly from the site. Admins receive, review, and manage messages through a dedicated backend panel to improve customer support and platform reliability.

1.5.5 Real-Time Logging and Admin Monitoring

A key feature that differentiates our platform is the integrated logging system. Admins can track user activities such as try-ons, interactions, and errors in real-time, offering deeper insights and better control over the platform's usage.

1.6 Methodology and Software Lifecycle

The development of Optical Facial Fit followed a modular, object-oriented, and incremental approach to ensure clarity, scalability, and efficiency.

1.6.1 Modular Development Approach

The system is broken into modules: user registration, AI try-on, product management, cart/favorites, contact form, logging, and admin panel. Each module was developed and tested independently for easier debugging and future scalability.

1.6.2 Object-Oriented Development (OOD)

OOD concepts were used to model real-world entities like User, Product, Tryon Session, and Feedback. This ensures the code is reusable, maintainable, and aligned with best practices.

1.6.3 Incremental Approach

Using the Incremental Model, features like registration and product browsing were implemented first, followed by try-on functionality, then admin tools. Each increment was tested before moving forward.

1.6.4 Rationale Behind the Selected Methodology

For the development of Optical Facial Fit, a hybrid approach combining modular, object-oriented, and incremental methodologies was selected to ensure efficient development, maintainability, and scalability.

This methodology ensures:

- **Modular Clarity:** Each component such as the AI virtual try-on, user management, admin panel, and product catalog was developed as a separate, self-contained module. This structure allowed focused development, easier debugging, and future feature expansion.
- **Code Reusability:** By implementing object-oriented programming (OOP), common functionalities such as user sessions, try-on logs, and product attributes were abstracted into reusable classes. This reduced code duplication and improved maintainability.
- **Continuous Testing:** Each module was tested independently after development. Incremental testing allowed for early identification of bugs, ensuring overall platform stability before integration.
- **Scalability:** The modular and OOP design enables the system to adapt quickly. New eyewear types, facial fitting algorithms, or even personalized AI recommendation engines can be integrated with minimal impact on the existing codebase.

1.6.5 Software Development Life Cycle (SDLC)

For Optical Facial Fit, the Incremental Model was adopted within the SDLC framework. This model divides the development process into smaller, manageable parts (increments), each of which delivers a functional component of the final system.

This approach ensured timely delivery of core features and allowed flexibility in refining or expanding the platform based on testing outcomes and user feedback.

1.6.5.1 Planning and Requirements Gathering

During this initial phase, detailed requirements were collected based on the core idea: delivering a user-friendly eyewear try-on platform powered by AI. The following key features were identified:

- AI-based virtual eyewear try-on (via camera)
- User registration, login, and profile customization
- Product management for different eyewear types
- Cart and favorites handling
- Admin dashboard for log management and customer interaction
- Real-time feedback and contact forms

1.6.5.2 Design

Based A detailed object-oriented design was created for each module:

- **Frontend Design:** A clean and responsive user interface using React.js and Bootstrap.
- **Backend Structure:** Modeled using OOP concepts such as User, Product, Try-On Session, and Admin Log.
- **Database Design:** Normalized MySQL schema using Laravel Eloquent ORM for relational data mapping.
- **AI Module:** Designed for facial landmark detection and virtual eyewear overlay.

These designs ensured the system components were logically connected yet individually manageable.

1.6.5.3 Implementation

Each Implementation followed an incremental rollout:

- Core modules like user authentication and product catalog were developed first.
- Cart, favorites, admin dashboard, and contact modules were added in later stages.

Each module was developed as a separate component, with testing and integration following completion.

1.6.5.4 Testing

After Multiple testing layers ensured the reliability of Optical Facial Fit:

- **Unit Testing:** Verified individual functions like adding items to the cart, facial landmark recognition, and login validation.
- **Integration Testing:** Ensured smooth communication between frontend, backend, and AI modules.
- **User Acceptance Testing (UAT):** Real users tested the virtual try-on experience, navigation, and responsiveness.

1.6.5.5 Deployment and Maintenance

Each increment was deployed to a staging environment for live testing. Once approved, it was deployed to the production server.

- Deployment involved configuring the Laravel backend, setting up databases, and hosting the platform on a secure web server.
- Maintenance is ongoing, including:
 - Bug fixes based on user feedback

Chapter 2

PROBLEM DEFINITION

2 Problem Definition

The Optical Facial Fit project aims to solve a growing issue in online eyewear shopping: the uncertainty of visual fit and appearance. While e-commerce has revolutionized shopping convenience, it fails to replicate the physical trial experience needed for purchasing eyewear, a product where fit and look are critical. This project bridges that gap using artificial intelligence (AI), facial recognition, and web technologies to create a virtual eyewear try-on platform that simulates the real-world try-before-you-buy experience online [3].

2.1 Problem Statement

The main problem arises when customers shop for glasses online but are unsure whether the selected frames will suit their facial features. This often leads to indecision, dissatisfaction, returns, or abandoning the purchase altogether. Traditional eyewear e-commerce platforms do not offer a reliable way to visualize how the glasses will actually look on the user's face [4].

Optical Facial Fit solves this problem by integrating an AI-based virtual try-on system that uses a webcam or uploaded image to detect the user's face, align the eyewear model correctly, and render it in real-time. This ensures more informed decisions, enhances user confidence, and improves customer satisfaction ultimately increasing conversion rates and reducing return rates for online eyewear retailers.

2.2 Deliverables Requirements

2.2.1 Optical Facial Fit Platform

This is the main software product a web-based application built with modern frameworks (such as Laravel, JavaScript, and AI libraries). The platform offers:

- **Virtual Try-On Feature:** Users can try glasses virtually using their webcam or uploaded image.
- **Real-Time Facial Alignment:** Facial landmarks and proportions are calculated using AI to place eyewear accurately.
- **Product Browsing:** Users can browse, filter, and search different categories and styles of eyewear.
- **Cart and Favorites System:** Users can add preferred products to their cart or save them as favorites.
- **Payment Integration:** Secure checkout with payment gateways (e.g., Stripe, PayPal).

- **User Accounts:** Register, log in, and manage orders.
- **Admin Panel:** Manage product listings, monitor usage logs, view and respond to user contact forms.
- **FAQ and Contact Page:** Assists users with common questions and allows direct communication.

2.2.2 Documentation

User Documentation: This documentation is aimed at end-users of the Optical Facial Fit platform. It provides clear, step-by-step instructions for navigating the interface, trying on eyewear using the AI-based virtual try-on system, managing profiles, and completing purchases. Additionally, it explains how users can interact through chat functionality, add items to favorites or the cart, and manage their order history.

Technical Documentation: This document is tailored for developers and technical stakeholders. It outlines the overall architecture of the system, including the backend logic, frontend design components, database schema, and any third-party or AI integrations. A breakdown of the source code structure, naming conventions, and dependencies is provided, along with configuration steps for setting up the development and deployment environment. It outlines the overall architecture of the system, including the backend logic, frontend design components, database schema, and any third-party or AI integrations.

Admin Guide: The admin guide is created to support platform administrators in managing users, product content, and performing essential system tasks. It includes instructions on logging into the admin panel, uploading and categorizing new eyewear, managing user feedback from contact forms, reviewing system logs, and performing routine maintenance

2.2.3 Testing Reports

This section includes comprehensive documentation of all testing procedures and results gathered throughout the software development lifecycle. It consists of well-defined test cases for each functional module, detailing the inputs, expected outcomes, and actual outcomes. Reports from unit testing, integration testing, and user acceptance testing (UAT) are included.

2.2.4 Deployment Setup

The deployment section details the process of releasing your Optical Facial Fit website to a live server. It includes steps for configuring the production environment, setting up a web server and ensuring PHP and Laravel dependencies are properly installed. This section also covers database configuration, environment variables, and any necessary optimizations for performance and security. The deployment process ensures that the site is accessible to users, with proper monitoring.

2.3 Development Requirements

2.3.1 Technologies and Tools Frontend

Frontend

The frontend will be built using HTML, CSS, and JavaScript with React.js or Vue.js for interactivity. Bootstrap and Sass will ensure a responsive design, while Pusher will enable real-time communication, supporting features like live chat and notifications.

Backend

The backend of Optical Facial Fit will be developed using Laravel (PHP) due to its robust features, scalability, and support for secure and maintainable web applications. MySQL will be utilized for storing relational data such as user profiles, product listings, virtual try-on logs, cart and favorites information, and contact queries, ensuring efficient and structured data management. Third-party APIs will be integrated to support real-time functionalities, such as AI-based virtual try-on and payment gateway services. For authentication, OAuth 2.0 will be employed to enable secure social login options like Google, enhancing user convenience. To ensure strong security throughout the platform, the system will implement CSRF protection, input validation, encryption, and secure storage practices to protect both user data and application integrity.

2.3.2 Server and Hosting Requirements

Hostinger is used for deploying the Optical Facial Fit platform, offering reliable hosting services and straightforward deployment options. The domain for the website is also registered through Hostinger to maintain consistency and ease of management. The server is configured to handle concurrent users efficiently, supporting real-time operations such as the AI virtual try-on and user interactions. SSL certificates are implemented to ensure secure data transmission between users and the platform.

2.3.3 Performance and Scalability

Load balancing and caching mechanisms are implemented to ensure optimal performance, particularly during peak traffic periods when many users may be trying on eyewear virtually or browsing products. Database optimization strategies are used to efficiently manage large volumes of data, including product information, user interactions, try-on history, and contact queries. Real-time processing capabilities are integrated to support smooth AI try-on experiences and timely handling of user actions, all while maintaining minimal latency [5].

2.3.4 Security and Privacy

The platform implements strong user authentication methods to prevent unauthorized access and ensure secure login processes. Role-Based Access Control (RBAC) is used to define clear boundaries between regular users and administrators, allowing differentiated access to features such as product management, user logs, and contact queries. Sensitive user data, including passwords and personal information, is securely stored using hashing and encryption techniques. Additionally, the system follows GDPR-compliant data handling practices to uphold user privacy and build trust.

2.3.5 Development Milestones

The development milestones for Optical Facial Fit are structured into five distinct phases. Phase 1 involves gathering requirements, defining user stories, and designing initial wireframes for the user interface and AI try-on experience. In Phase 2, core features such as user authentication, product catalog, cart and favorites, and contact forms are developed. Phase 3 focuses on integrating real-time components, including the AI-based virtual try-on system and responsive admin panel for managing user activity and product data. Phase 4 includes comprehensive testing, bug fixing, and deployment of a beta version for initial user feedback. Finally, Phase 5 covers the full deployment of the platform, along with delivering user documentation and training resources to ensure a smooth onboarding experience.

2.3.6 Collaboration and Version Control

Git and GitHub are used for source code management and version tracking, ensuring organized development, seamless collaboration, and efficient tracking of changes throughout the lifecycle of the Optical Facial Fit project.

2.4 Current System

At present, there is no single eyewear e-commerce platform that fully integrates the features necessary for a comprehensive, interactive, and user-friendly virtual try-on experience like the one developed in Optical Facial Fit. Existing systems, such as standard online optical stores and e-commerce platforms, provide basic services like product listings and standard checkout processes, but they lack the integration of advanced features such as AI-driven virtual try-on, personalized suggestions, real-time product interaction, and efficient admin-user communication. Most current platforms focus on isolated components like static product displays or basic customer support, but do not combine these into a cohesive system that supports immersive shopping, real-time AI experiences, and effective user engagement [6].

Typical eyewear platforms often offer static content such as product images or videos, which fall short of delivering the interactivity needed for users to make confident purchase decisions. Furthermore, many of them lack advanced scalability, personalization options, and efficient management tools for both users and administrators. Customer support features are usually limited to email or delayed responses, missing real-time engagement options that can improve satisfaction and reduce abandonment rates.

In contrast, Optical Facial Fit aims to combine the best elements of these systems while resolving their key limitations. It provides a dynamic, AI-powered virtual try-on experience, an intuitive and real-time interface, robust admin tools, and personalized user features such as favorites, cart management, and quick product access all designed to create a seamless, modern, and engaging eyewear shopping experience.

Chapter 3

REQUIREMENT ANALYSIS

3 Requirement Analysis

The process of creating a use case diagram for Optical Facial Fit involves a structured approach beginning with stakeholder analysis to understand the needs of users, administrators, and other involved parties. This is followed by identifying and describing key use cases such as virtual try-on, product browsing, cart management, contact queries, and admin functionalities. Relationships among these use cases are then established to show how different users interact with the system. Finally, the use case diagram is constructed to provide a clear visualization of system interactions.

This method ensures that all functional requirements are well-defined and aligned with stakeholder expectations, offering a precise blueprint to guide developers in designing and implementing the platform effectively [7].

3.1 Use Case

The Optical Facial Fit use case diagram illustrates the interaction between two primary actors: the Customer and the Administrator. This use case represents a comprehensive overview of the functionalities available on the platform for both user roles. Customers can register or log in, browse and view eyewear products, utilize AI-powered virtual try-on technology to visualize glasses on their face, receive personalized AI recommendations, and proceed through the e-commerce flow by adding items to their cart, checking out, and managing reviews.

On the other hand, Administrators are responsible for maintaining the system by managing products, handling users, monitoring platform health, addressing support queries, and configuring AI modules, including overseeing the AI virtual try-on engine. The system is designed to offer a seamless and interactive experience for customers, leveraging real-time AI features while giving administrators full control over platform functionality and data integrity.

This use case serves as a blueprint for the system's core operations, ensuring both functional completeness and user engagement by integrating AI with traditional e-commerce components.

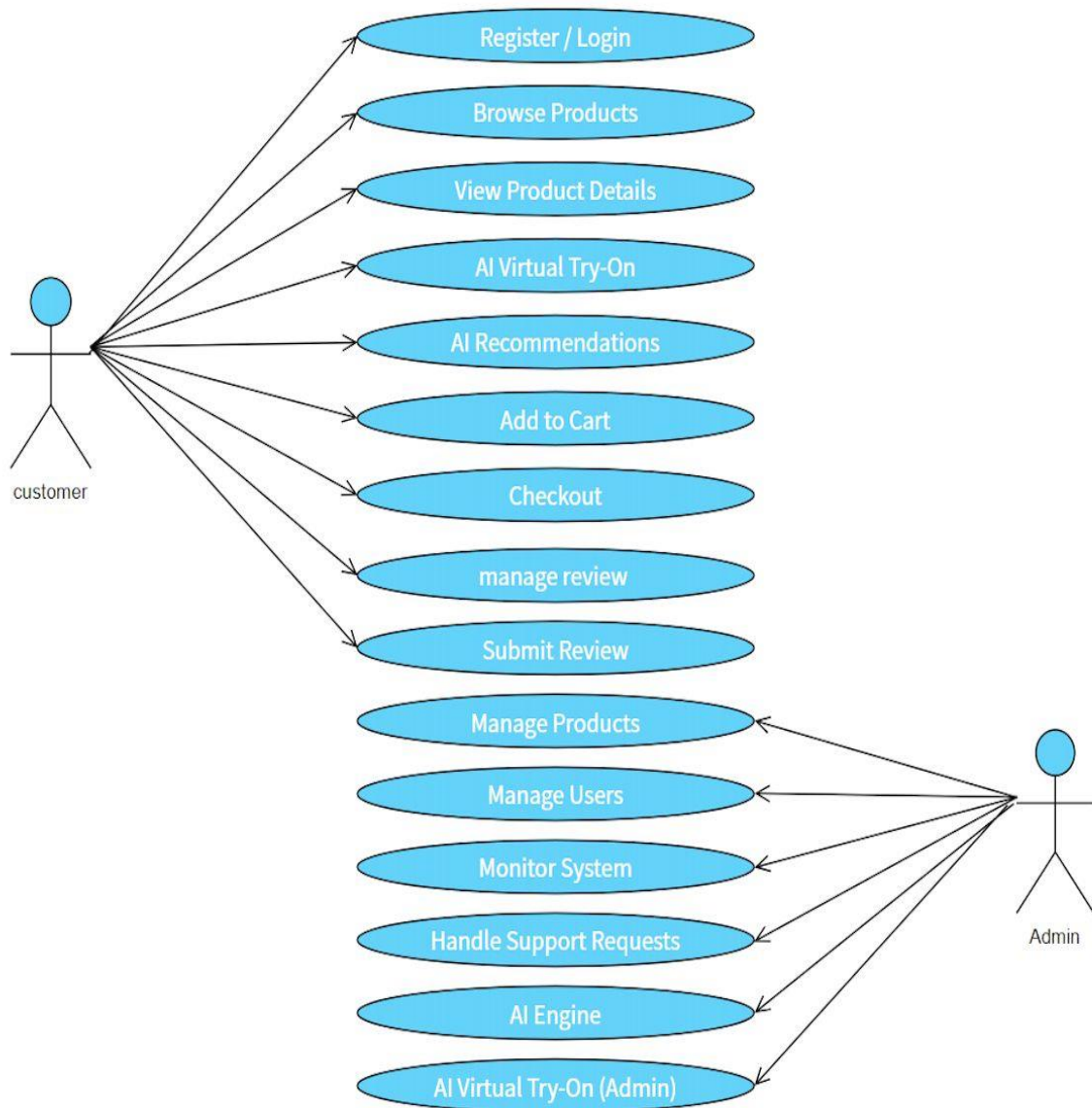


Figure 3.1-1use case

3.1.1 Use Case Description

This use case diagram illustrates the interaction between two primary actors: the Customer (Registered User) and the Administrator within the Optical Facial Fit platform.

Customers can register/log in to the platform, browse available eyewear products, view product information, and use the AI-powered Virtual Try-On feature to preview eyewear on their face in real-time. Customers can add products to their cart, proceed to checkout, and submit reviews.

Administrators have elevated privileges that allow them to manage products, handle user data, monitor the system, and respond to support queries.

3.1.1.1 Complete Platform Interaction

The Complete Platform Interaction Table provides a comprehensive overview of all possible interactions between system actors (Customer and Admin) and the platform.

Table 3-1 complete platform interaction

Use Case ID	UC
Use Case Name	Try On Eyewear Using AI
Actors	Primary Actor: Registered User Secondary Actor: Visitor (limited preview access), Administrator (for monitoring usage)
Description	This use case describes how a user interacts with the AI-based virtual try-on system to preview eyewear on their face in real-time before making a purchase decision.
Trigger	The user clicks the “Try On” button associated with a product on the website.
Preconditions	The user has access to a device with a camera. The AI module is active and properly integrated. The eyewear product is available and tagged for virtual try-on.
Postconditions	The user is able to see a real-time preview of how the eyewear looks on their face. The session data may be stored for analytics or personalized suggestions.
Normal Flow	Access Try-On Feature • User logs in (optional for preview). • Navigates to a product page. • Clicks on the “Try On” button. Camera Access & AI Detection • System requests camera permission. • User grants access. • AI model loads and detects facial features. Display Output • Eyewear is overlaid on the user’s face in real-time. • User can switch between models to try different products
Alternative Flows	Camera Denied: If the user denies camera access, a message is generated to allow camera access for real time try on.
Exceptions	Poor lighting or camera quality may prevent accurate overlay. Unsupported browsers may block access to the AI module.

3.1.1.2 Normal flow

The Normal Flow Table defines the standard and most commonly followed sequence of steps taken by users (both customers and admins) when interacting with the platform. It includes the usual behavior patterns such as account registration, product browsing, using the AI try-on feature, managing the cart, and submitting reviews for customers, as well as day-to-day administrative actions. This table is essential for identifying the expected operational path under regular usage without exceptions or errors.

Table 3-2 normal flow

Step No.	Customer Actions	System Response
1	Registers/Login to the platform	Validates credentials and grants access
2	Browses products	Displays product catalog with filters
3	Views product details	Loads product description, pricing, images
4	Uses AI Virtual Try-On	Activates camera/face overlay and renders virtual glasses
5	Receives AI Recommendations	Fetches suggestions based on facial features and preferences
6	Adds product to cart	Stores product data in cart
7	Proceeds to checkout	Displays billing and payment form
8	Completes payment	Processes transaction and stores order details

Step No.	Customer Actions	System Response
9	Submits a review	Saves and displays the review on the product page

3.1.1.3 Admin Flow

The Admin Workflow refers to the sequence of operations carried out by an administrator to manage and maintain the Optical Facial Fit platform. This includes product and user management, AI module supervision, system monitoring, and customer support handling. The workflow ensures the smooth operation of both customer-facing services and backend functionalities.

Table 3-3 Admin flow

Step No.	Admin Actions	System Response
A1	Logs into the admin panel	Verifies admin access and shows dashboard
A2	Manages products (add/edit/delete)	Updates product database
A3	Manages user accounts	Enables/disables or updates user data
A4	Monitors system and reviews logs	Displays server health, AI logs, error reports
A5	Handles support requests	Retrieves and updates ticket statuses

3.2 Functional Requirements

These are the following modules that will be included in this system.

- i. Customers Modules
- ii. Admin Modules

3.2.1 Customer's Module

- Customers can register and log in using email and password.
- Email verification and password recovery are available.
- Customers can upload and update profile pictures.
- Users can update personal information (name, contact details, preferences).
- Customers can upload a photo for facial analysis.
- The system detects facial features and determines face shape.
- Personalized glasses recommendations are provided.
- Recommendations can be filtered and sorted by brand, price, and style.
- Customers can add items to a shopping cart.
- Secure checkout and order tracking are available.

3.2.2 Admin modules

- Admins can manage user accounts (approve, delete, update).
- Admins can add, update, and remove products from the catalog.
- Monitor and analyze facial recognition performance.
- Manage encryption settings and compliance with data protection regulations.
- Review and address customer feedback and inquiries.
- Monitor website performance and manage inventory.

3.3 Non-functional requirement

Non-functional requirements are critical to ensuring that our online glasses shopping platform performs well, is user-friendly, secure, reliable, and accessible across different platforms. These requirements complement the functional aspects of the project by addressing key performance and quality attributes. Some of the non-functional requirements for auto shipping management system are given below.

3.3.1 Usability

Our system must have an intuitive and user-friendly interface, allowing users to easily navigate through the website, upload photos, and receive personalized recommendations. The website should be accessible to users with disabilities, ensuring an inclusive experience for all users. Additionally, the design must be responsive to provide optimal functionality and appearance on various devices, including desktops, tablets, and smartphones, ensuring a consistent and enjoyable user experience regardless of the device used.

3.3.2 Performance

Performance is a key aspect of our system, which should ensure that the website loads within 2 seconds on a standard broadband connection to provide a seamless user experience. The system must be scalable to handle a growing number of users and data without significant performance degradation. Furthermore, facial recognition and image analysis should be completed within 5 seconds to maintain user engagement and ensure that the process is efficient and effective.

3.3.3 Security

Security is paramount for protecting user data and maintaining trust in our platform. User data, including photos and personal information, must be stored securely using encryption techniques to prevent unauthorized access. Our system must comply with data protection regulations such as GDPR and CCPA to ensure that user privacy is respected, and legal requirements are met. Additionally, all financial transactions must be secured using industry-standard encryption protocols like SSL/TLS to protect user data during payment processes, ensuring that transactions are safe and secure.

3.3.4 Portability

Our system should be designed to be portable, allowing it to operate across different environments and platforms without requiring significant changes. This includes compatibility with various operating systems and web browsers to ensure that a wide range of users can access the system without any issues. The code base should be modular and well-documented to facilitate easy updates and migrations to new environments, ensuring long-term sustainability and adaptability.

3.3.5 Reliability

While the platform primarily relies on a structured relational database for managing core data such as user accounts, product listings, orders, and reviews, it also incorporates semi-structured data formats like JSON in specific functional areas.

Chapter 4

DESIGN AND ARCHITECTURE

4 Design and Architecture

The design and architecture of Optical Facial Fit are centered around developing a modular, scalable, and interactive e-commerce platform specialized for eyewear with AI-powered virtual try-on capabilities. The frontend ensures a responsive and intuitive user experience for browsing products, using the virtual try-on feature, and completing purchases. The backend, built using Laravel (PHP), is responsible for secure data handling, real-time interactions, and well-structured module integration, including product management, user activity logging, and AI recommendation systems. This cohesive architecture supports smooth operations, intelligent product suggestions, and personalized shopping experiences, enhancing both customer satisfaction and administrative efficiency. The backend, built using Laravel (PHP), is responsible for secure data handling, real-time interactions, and well-structured module integration, including product management, user activity logging, and AI recommendation systems [8].

4.1 System Architecture

Diagram 4.1 illustrates how Optical Facial Fit serves users through a web-based platform that integrates e-commerce functionalities with AI-powered virtual try-on technology. At its core, the system supports essential features such as user registration/login, product browsing, detailed product views, AI virtual try-on, cart management, secure checkout, and user reviews. For administrators, it includes modules for product and user management, system monitoring, and AI engine control.

Users (customers) interact with the platform through intuitive web interfaces, accessing features like virtual try-on and personalized recommendations. The backend powered by Laravel (PHP), processes these interactions and communicates with third-party APIs, AI services, and the database for real-time processing and secure data management. All transactions and activities are routed through a central server that ensures real-time performance, scalability, and data security, allowing seamless shopping and efficient admin operations. . The backend powered by Laravel (PHP), processes these interactions and communicates with third-party APIs, AI services, and the database for real-time processing and secure data management

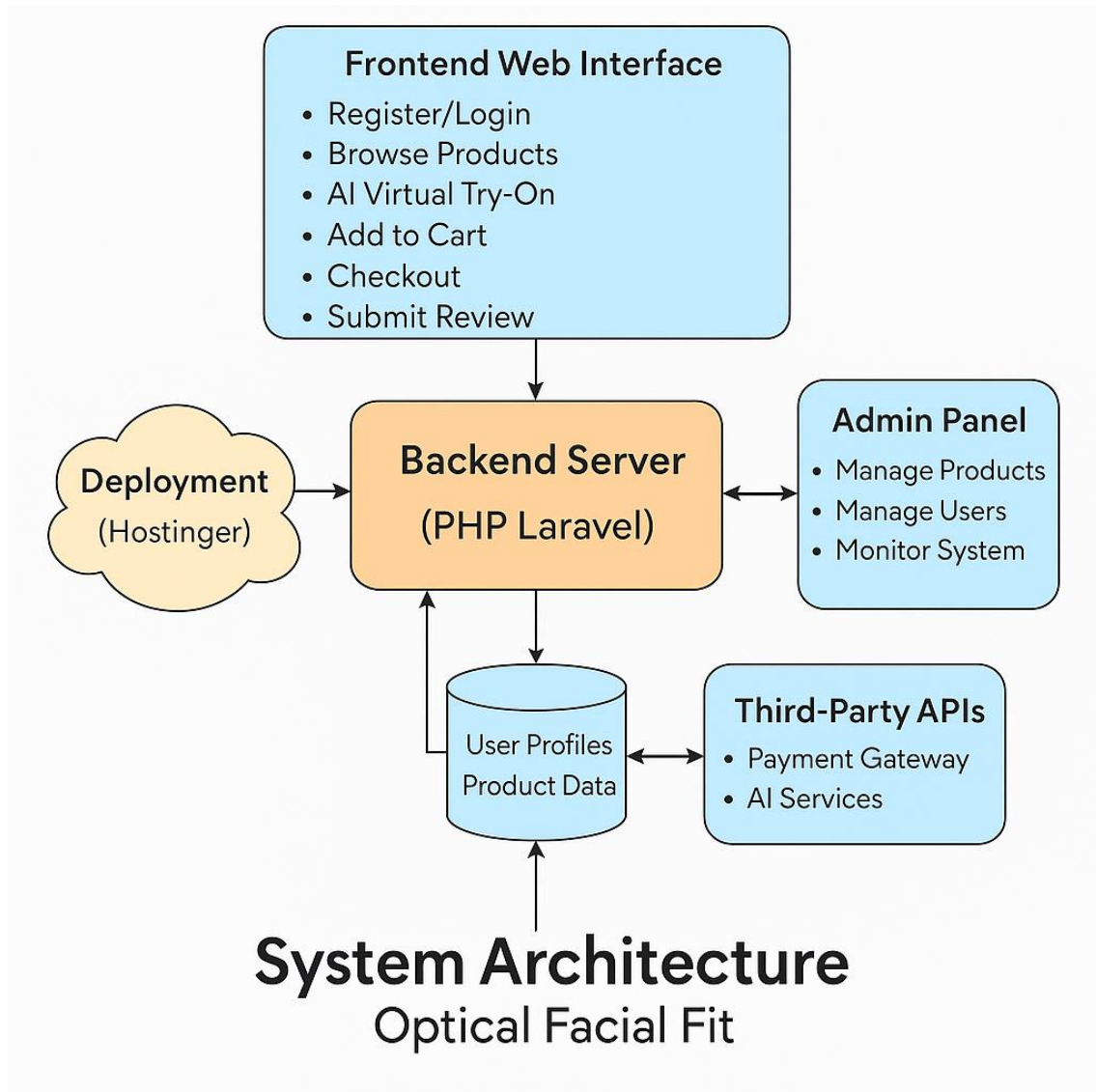


Figure 4.1-1 System architecture

The system architecture of our AI Try-On Eyewear Platform is designed to provide a seamless and immersive user experience by integrating virtual try-on technology with intuitive browsing and secure ordering. The architecture follows a three-tiered structure: the Frontend Layer, the Backend Layer, and the Database Layer.

The **Frontend Layer** acts as the primary user interface, allowing visitors and registered users to interact with the platform. It includes modules such as the AI-based virtual try-on system for glasses, product galleries, user authentication interfaces, shopping cart, order tracking, and personalized recommendations all designed to be responsive and user-friendly across devices.

The **Backend Layer** handles the core application logic and services. It processes user inputs from the try-on interface, manages product data, handles user authentication and order placement, integrates with the AI engine for real-time try-on rendering, and manages interactions like reviews and recommendations. This layer ensures secure and efficient communication between the frontend and the database, enforcing all business rules and application workflows.

The **Database Layer** securely manages and stores data such as user profiles, product catalogs, order histories, try-on session data, and reviews. Each component in the backend interacts with structured database entities to maintain data integrity, optimize performance, and provide real-time updates across the platform.

4.2 Data Representation

This architecture ensures data integrity, consistency, and high performance across all interconnected modules, including virtual try-on sessions, product management, user accounts, and order processing.

Given the interactive and feature-rich nature of the platform which includes AI-driven try-on capabilities, user registration, product browsing, shopping cart functionality, real-time order tracking, and customer reviews the system relies on predefined database schemas to manage data efficiently across various components.

Each core entity, such as users, glasses frames, try-on data, orders, and reviews, is stored in well-structured tables with clear primary and foreign key relationships. This structure allows the system to accurately map user interactions with products and services, maintain consistent data flow, and ensure reliability during real-time operations [9].

The platform utilizes Structured Query Language (SQL) to execute complex queries, data joins, filters, and aggregations all of which are essential for dynamic search, personalized recommendations, and detailed performance analytics. This robust data layer is key to delivering a fast, secure, and personalized shopping experience.

While our AI Try-On Eyewear Platform primarily relies on a structured relational database model, it also incorporates semi-structured formats like JSON in specific areas. These include storing dynamic AI-generated try-on data, product customization preferences, real-time user interaction logs, and API responses from external services such as payment gateways or shipping providers.

This hybrid approach balances the reliability and efficiency of SQL with the flexibility and adaptability of lightweight data formats. JSON is particularly useful in managing variable data

structures such as virtual try-on session metadata or logs of user facial landmarks, which can vary in structure and size.

By combining structured and semi-structured data models, the platform ensures robust performance, enhanced scalability, and the ability to adapt quickly to evolving features all while maintaining a seamless and responsive user experience across the entire system.

Optical Facial Fit – Data Representation

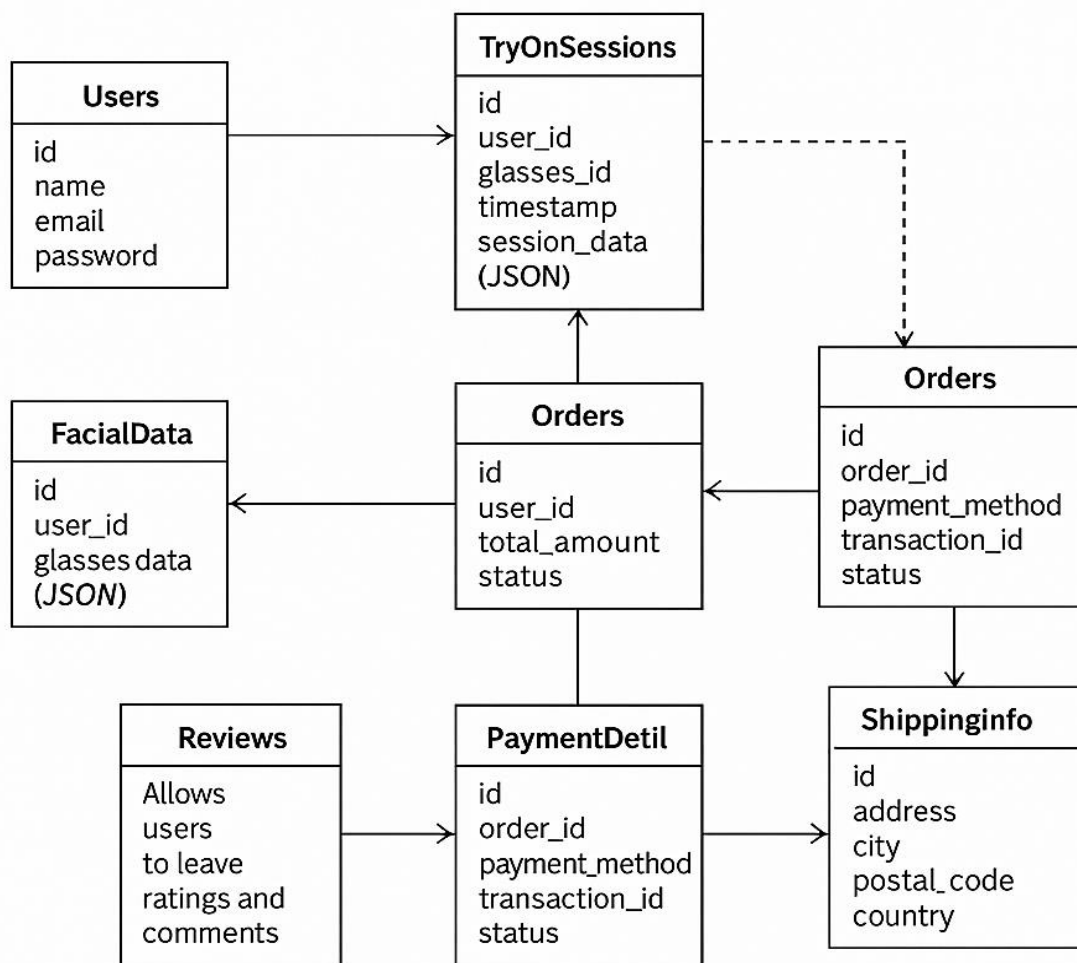


Figure 4.2-1 Data representation

4.3 Process Flow/Representation

The Optical Facial Fit (OFF) platform process flow begins with user registration and authentication, providing secure access to features such as product browsing, AI-based virtual try-on, and smart

recommendations. Once logged in, users can explore eyewear products, visualize them using the virtual try-on tool, receive personalized AI suggestions, and add items to their cart. The system ensures smooth data flow between the frontend and backend, supporting real-time product rendering, secure checkout, notifications, and seamless management of user interactions, reviews, and purchases.

4.3.1 Activity Diagrams

Activity diagrams play a vital role in modeling the dynamic behavior of a system, offering a comprehensive visual representation of how various actions flow and interact. These diagrams depict sequential, branching, and parallel workflows, using constructs such as decisions, forks, joins, and transitions to represent concurrent and conditional activities effectively.

They are especially useful for illustrating how different components within a system coordinate and evolve over time, making complex processes easier to understand. This clarity enhances system analysis, design, and stakeholder communication, ensuring that all parties involved share a consistent understanding of the system's behavior [10].

Moreover, activity diagrams help identify inefficiencies, bottlenecks, or redundancies in workflows. By visualizing all possible action flows, they enable better process optimization and facilitate forecasting the impact of changes on system performance. During the planning and refinement phases, activity diagrams serve as proactive tools for refining operations, supporting smoother and more efficient system implementation.

4.3.2 Admin Activity Diagram

The Admin Activity Diagram illustrates the essential tasks performed by the administrator within the Optical Facial Fit platform. It begins with the admin logging into the system and accessing the admin dashboard. From there, the admin can:

1. **Manage Users:** View, edit, or remove user accounts and monitor user activity.
2. **Upload & Manage Glasses Inventory:** Add new eyewear products, update product details (e.g., style, price, availability), and remove outdated items.
3. **Review Try-On Analytics:** Monitor session data, facial fit results, and user preferences gathered from AI try-ons.
4. **View Orders:** Access user orders, check status, and resolve issues such as payment failures or returns.

5. **Manage Reviews:** Approve or remove user reviews to maintain quality and authenticity.
6. **Track Payments & Shipping:** Oversee transactions, verify successful payments, and ensure timely dispatch of products.

The diagram ensures a streamlined flow from login to specific administrative operations, reinforcing system oversight and ensuring a high-quality user experience.

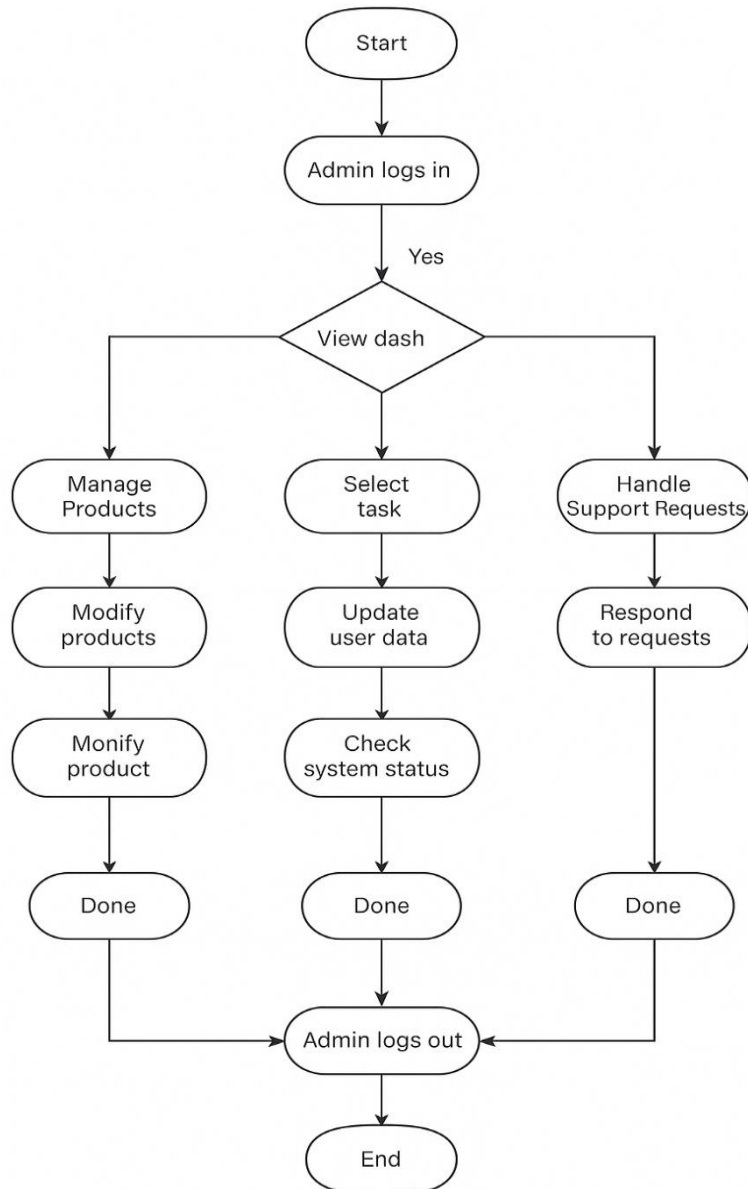


Figure 4.3-1 Admin activity

4.4 Process Flow

A Process Flow Diagram (PFD) is a visual representation that maps out the sequential steps, decisions, and interactions within a system or workflow. It highlights how inputs move through various stages of processing, showing the relationships and data exchange between users, system components, and external services. The purpose of a PFD is to provide a clear, high-level overview of the entire operation, enabling stakeholders to understand how the system functions as a whole and identify any potential inefficiencies or bottlenecks. The purpose of a PFD is to provide a clear, high-level overview of the entire operation, enabling stakeholders to understand how the system functions as a whole and identify any potential inefficiencies or bottlenecks.

Here's the Process Flow Diagram (PFD) for the "Optical Facial Fit" project, which clearly outlines the detailed sequence of user interactions and system operations that deliver the virtual try-on experience. This diagram illustrates how the user initiates actions such as logging in or registering, uploads or captures their facial image, and how this input flows through the AI-powered facial recognition and analysis modules. It further depicts how the system processes this data to match and display suitable eyewear options in real-time. Additionally, the flow shows the interaction between the user interface, backend services, the product catalog, shopping cart management, and the payment system, highlighting how these components seamlessly coordinate to provide a smooth and engaging user experience. The diagram provides a comprehensive overview of the end-to-end process, from initial user engagement to completing a purchase, ensuring clarity on how each module contributes to the platform's functionality. It further depicts how the system processes this data to match and display suitable eyewear options in real-time. Additionally, the flow shows the interaction between the user interface, backend services, the product catalog, shopping cart management, and the payment system, highlighting how these components seamlessly coordinate to provide a smooth and engaging user experience. The complete process flow diagram is as follows:

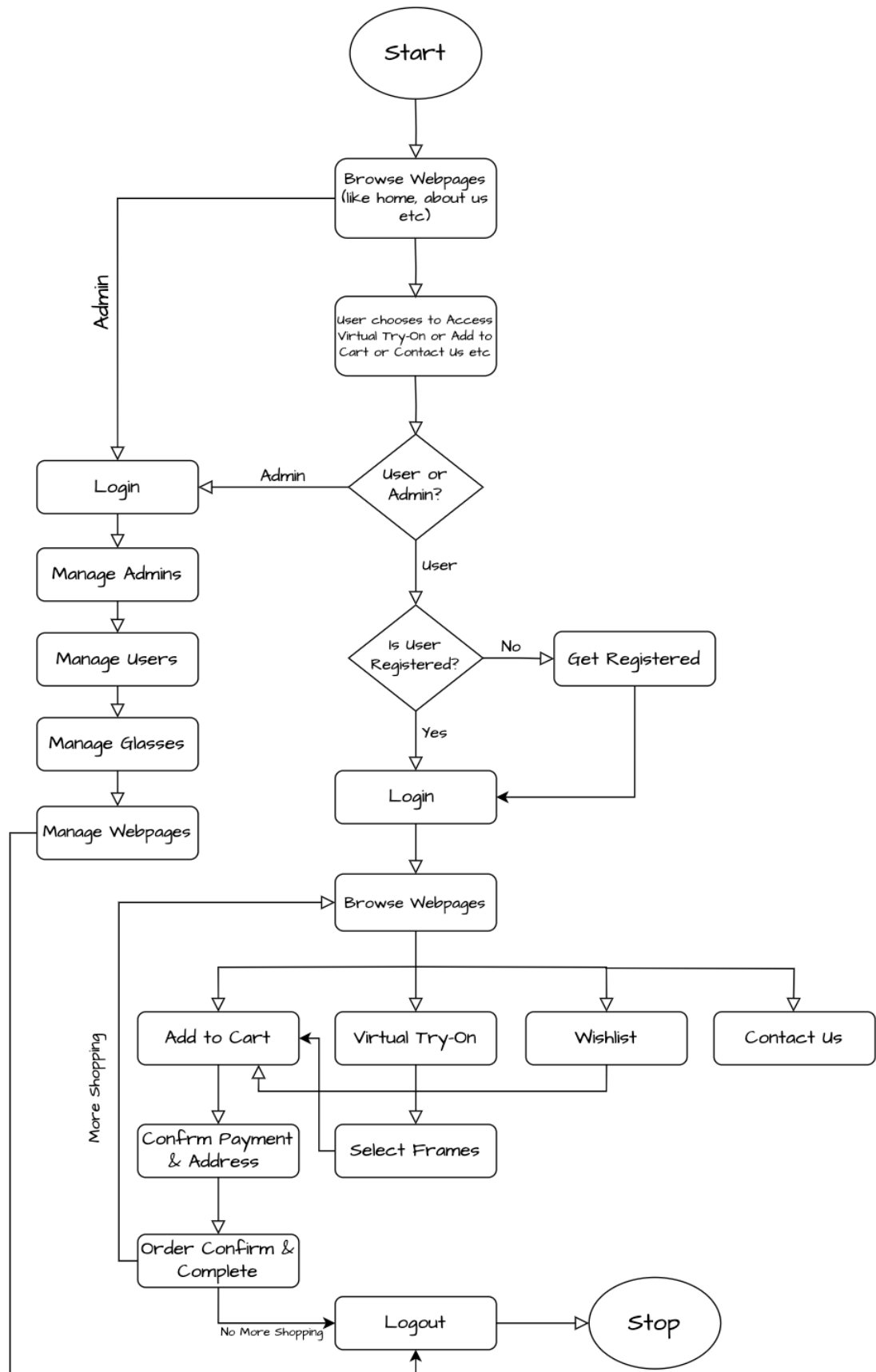


Figure 4.4-1 Process flow

4.5 Design Models

The design model of Optical Facial Fit (OFF) provides a structured blueprint outlining how the system's core components such as user management, virtual try-on powered by computer vision, product catalog, and admin panel interact to deliver a seamless and intuitive eyewear shopping experience. It emphasizes a user-centric interface that allows easy navigation and real-time virtual try-on through facial landmark detection. The model details the data flow between the frontend and backend, ensuring smooth integration of Laravel for backend processing, MySQL for data storage, and CV algorithms for accurate frame fitting. Security measures for authentication and scalability considerations are embedded to support future growth, making the platform reliable, efficient, and engaging for users and administrators alike.

4.5.1 Sequence Diagram

The Sequence Diagram of the "Optical Facial Fit" project comprehensively illustrates the detailed interaction flow between the user and the system's key components, capturing the essential functionalities of the platform. It outlines each step the user takes, starting from accessing the website, through authentication processes such as login or registration, to engaging with the virtual try-on feature powered by AI-based facial analysis using computer vision. The diagram further depicts browsing the eyewear catalog, selecting frames, adding items to the shopping cart, and proceeding to checkout. Crucially, it shows the coordinated communication between the user interface, backend services including authentication modules, the facial recognition and fitting algorithms, and the recommendation engine and the database for storing user profiles, product details, and transaction records. By mapping these interactions in sequence, the diagram provides a clear blueprint of the system's dynamic behavior, ensuring that all processes are logically ordered and optimized to meet user needs effectively and efficiently. The sequence diagram of our project is as follows:

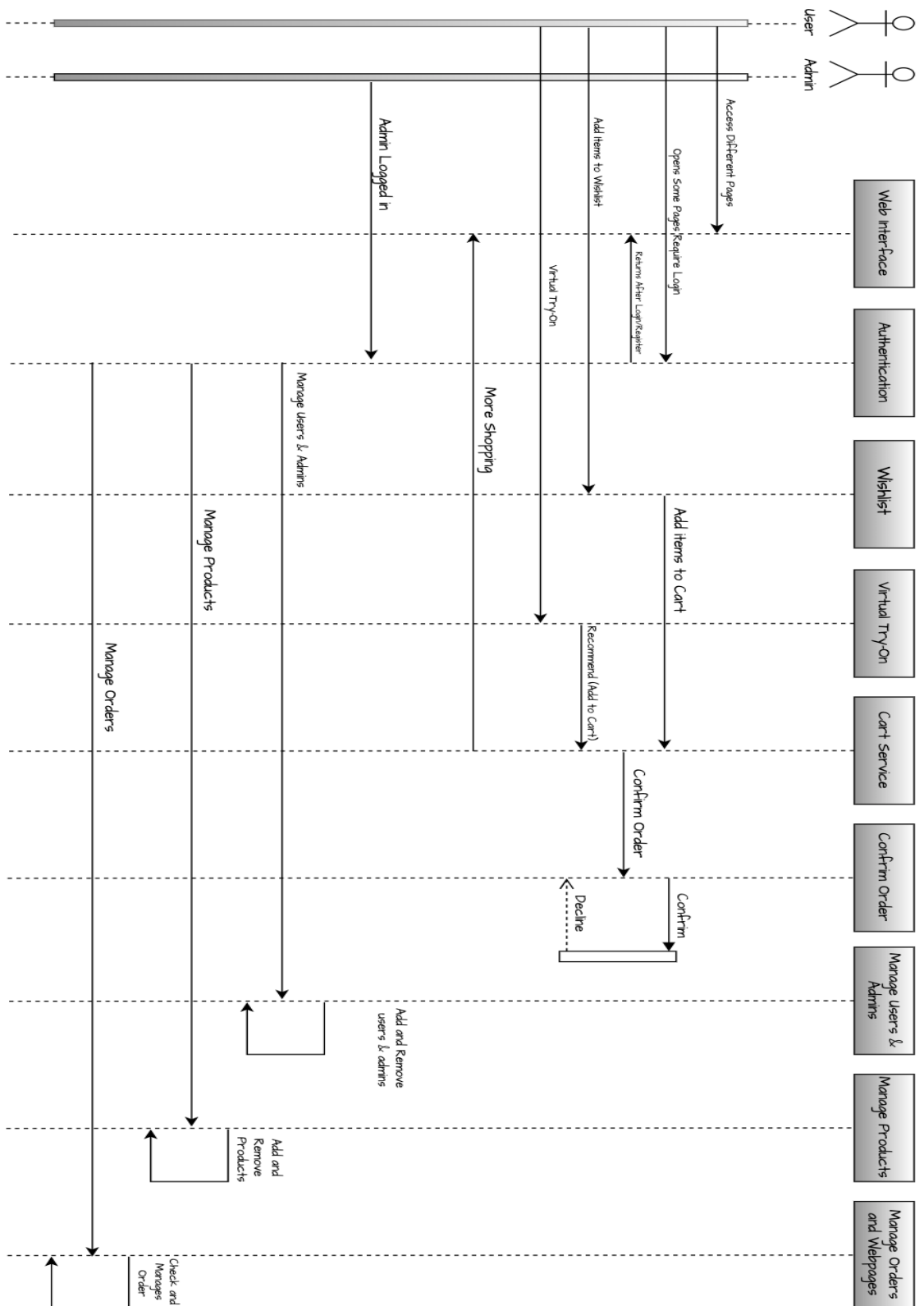


Figure 4.5-1 Sequence Diagram

4.5.2 Class Diagram

A **Class Diagram** is a type of **UML (Unified Modeling Language)** diagram used to represent the static structure of a system by showing its classes, attributes, methods (functions), and the relationships between objects. It is a foundational blueprint in object-oriented design and is especially useful for understanding how different parts of a system interact at a structural level.

Class diagrams provide a clear map of:

- What data each class holds (attributes),
- What actions each class performs (methods),
- How classes are connected (associations, inheritance, dependencies),
- And how data flows or is grouped across the system (aggregation, composition).

The Class Diagram for the "Optical Facial Fit" project provides a detailed visualization of the system's static architecture by illustrating its core classes, attributes, methods, and the relationships among them. It encapsulates the structure of key entities such as User, Admin, Product, Order, Cart, Wishlist, and Review. Each class defines specific attributes (e.g., User has name, email, password; Product has title, price, stock) and methods (e.g., login(), addToCart(), processPayment()) relevant to its functionality. The diagram showcases inheritance (e.g., Admin inherits from User), associations (e.g., a User can place multiple Orders), and aggregations (e.g., an Order aggregates multiple Products). Additionally, it represents one-to-many and many-to-many relationships where applicable, such as between users and their Wishlist or Cart items.

This class diagram serves as a blueprint for the database schema and object-oriented programming logic. It ensures consistency across development, facilitates code maintenance, and provides a clear understanding of the system's components and their interconnections.). Additionally, it represents one-to-many and many-to-many relationships where applicable, such as between users and their Wishlist or Cart items.

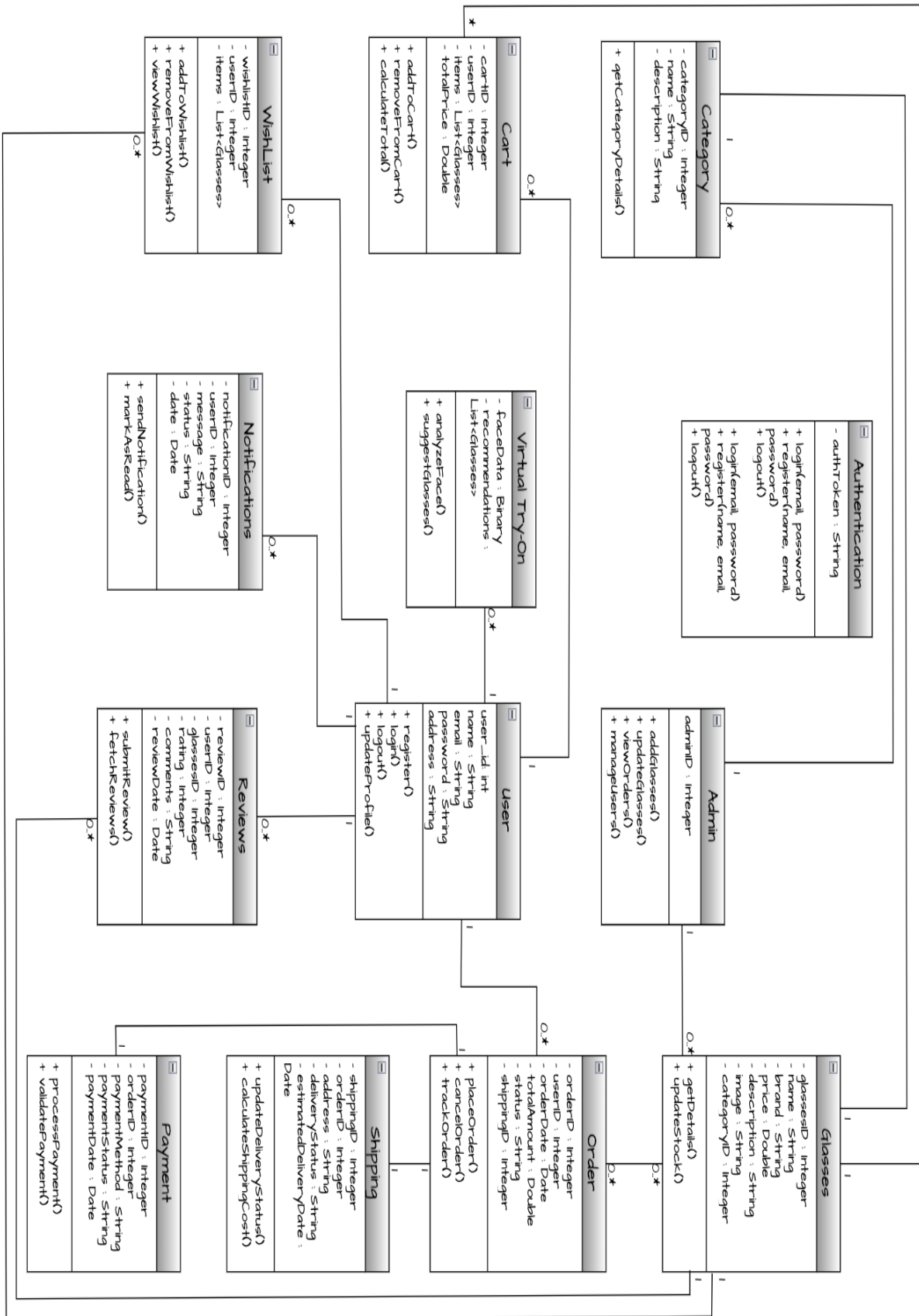


Figure 4.5-2 Class diagram

4.6 Data Design

As we discussed all the aspects of Optical Facial Fit, now we will discuss the data design of our project. It describes the whole data structure of our application to ensure an efficient and secure system.

- **User Information** will store details about users, including their names, contact information, passwords (securely encrypted), and addresses. It will also include user roles (admin or customer) to manage permissions effectively.
- **Admin Information** will maintain details about administrators managing the application, including credentials and activity logs to monitor system interactions.
- **Glasses Information** will store data about the glasses available in the system, such as product ID, name, brand, price, stock quantity, category, description, and images.
- **Virtual Try-On Data** will handle temporary data generated during the virtual try-on process, ensuring real-time performance while keeping privacy intact.
- **Cart Information** will include details about items added to a user's cart, such as product IDs, quantities, and the calculated total price.
- **Order Information** will maintain records of user orders, including order ID, user ID, product details, total price, shipping address, and order status.
- **Payment Information** will securely store data related to user transactions, including payment method, transaction ID, and billing details.
- **Category Information** will classify glasses into categories for easy browsing, storing category names and descriptions.
- **Database Security measures**, such as user authentication and secure data storage, will be implemented to protect sensitive information.

4.6.1 Data Dictionary

The Data Dictionary for the "Optical Facial Fit" project is a list of the names and attributes used within the system. These elements define the key components that will be implemented in the project. It serves as a reference for developers, designers, and stakeholders to ensure consistency in data usage and management. By standardizing data definitions, it helps streamline the development process and improve system efficiency. The dictionary includes details such as data types, formats, and relationships between different entities in the system.

- **User:** A person who visits the website to try on glasses and make a purchase.
Attributes: Name, Contact Information (email, phone), Preferences (frame style, lens type), Login Credentials (username, password).
- **Facial Data:** Data captured through the user's camera to analyze their face for personalized glasses recommendations.
Attributes: Face Shape, Eye Distance, Nose Width, Chin Position, Eye Position, Other Facial Features (collected via AI facial recognition).
- **Glasses:** Information about the glasses available for virtual try-on and purchase.
Attributes: Frame Style, Frame Material, Lens Type, Lens Color, Frame Color, Size, Price, Image.
- **Try-On Session:** A record of the user's interaction with the virtual try-on feature, storing details of glasses tried on.
Attributes: User ID, Glasses ID, Date/Time, Selected Frame Style, Selected Lens Type, Virtual Try-On Image.
- **Order:** Details of a user's purchase and order history.
Attributes: Order ID, User ID, Glasses ID, Quantity, Total Price, Shipping Address, Order Status, Payment Status, Order Date.
- **Payment:** Information about the financial transaction associated with a user's purchase.
Attributes: Payment ID, Order ID, Payment Method (credit card, PayPal, etc.), Transaction ID, Payment Status, Payment Amount, Payment Date.
- **Shipping:** Information related to the delivery of glasses after an order is placed.
Attributes: Shipping ID, Order ID, Shipping Address, Shipping Method, Shipping, Estimated Delivery Date, Tracking Number, Shipping Date
- **Cart:** A temporary storage of items that a user intends to purchase.
Attributes: Cart ID, User ID, Glasses ID, Quantity, Date Added, Total Price, Cart Status
- **Wishlist:** A collection of glasses that a user is interested in but has not yet added to their cart.
Attributes: Wishlist ID, User ID, Glasses ID, Date Added, Wishlist Status.

Chapter 5

IMPLEMENTATION

5 Implementation

The implementation phase of the Optical Facial Fit (OFF) platform translates system design into a fully operational web-based eyewear e-commerce solution. This stage involves the practical construction of core functionalities including the AI-powered virtual try-on, secure user management, product browsing, recommendation systems, and seamless checkout processes. The development process is modular and iterative, ensuring that each component integrates smoothly into the broader platform.

Laravel is chosen as the backend framework due to its MVC architecture, security features, and extensive support for RESTful APIs. The frontend is built using HTML, CSS, JavaScript, and Blade templates to create an intuitive, responsive interface suitable for all device sizes. Vue.js is also incorporated for reactive UI elements. The platform supports dynamic interactions such as live AI try-on, personalized recommendations, and real-time admin management.

Key modules include user authentication, product catalog handling, virtual try-on using webcam access and AI overlays, intelligent recommendations, cart management, checkout via payment gateways, and admin capabilities like product/user oversight and system monitoring. APIs and libraries are integrated for tasks such as AI engine communication, cloud-based image handling, real-time notifications, and secure email delivery.

5.1 Algorithms

5.1.1 User Authentication

Function: Manages registration and login for users and admins.

Input: Email, password

Output: Authenticated session

Pseudocode:

- BEGIN
- IF login form submitted THEN
- VALIDATE email and password
- CHECK user record in database
- IF valid THEN

- START user session
- ELSE
- DISPLAY error
- ENDIF
- END

5.1.2 AI-Based Virtual Try-On

Function: Allows users to virtually try glasses using their camera or uploaded photo.

Input: Facial image, product ID

Output: Rendered image with selected glasses

Pseudocode:

- BEGIN
- IF user initiates try-on THEN
- CAPTURE/UPLOAD user facial image
- DETECT facial landmarks
- OVERLAY selected eyewear based on alignment
- DISPLAY try-on preview
- END

5.1.3 Cart & Checkout System

Function: Manages product selection, cart updates, and order placement.

Input: Product ID, quantity

Output: Order details and confirmation

Pseudocode:

- BEGIN
- IF user adds product to cart THEN
- UPDATE cart with product ID and quantity
- IF user proceeds to checkout THEN
- VALIDATE cart items and user data
- PROCESS payment via gateway

- CONFIRM order and notify user
- END

5.1.4 Review Management

Function: Allows users to post and manage product reviews.

Input: Review text, rating

Output: Stored user feedback

Pseudocode:

- BEGIN
- IF user submits review THEN
- VALIDATE input
- SAVE review with timestamp and user ID
- DISPLAY updated reviews
- END

5.1.5 Admin Product & User Management

Function: Enables admin to manage inventory and users.

Input: Admin panel actions (add/edit/delete)

Output: Updated database records

Pseudocode:

- BEGIN
- IF admin logs in THEN
- GRANT access to dashboard
- ALLOW product/user CRUD operations
- UPDATE database accordingly
- END

5.2 External APIs

The Optical Facial Fit (OFF) platform leverages select technologies to deliver an intuitive and interactive eyewear try-on experience without relying on third-party APIs like Google or GitHub.

The platform is built using PHP Laravel and utilizes computer vision algorithms for AI-based facial landmark detection and frame fitting. This allows users to virtually try on glasses in real time using their webcam or uploaded images, handled via internal controllers such as Try On Controller and services responsible for facial analysis.

Although OFF does not support real-time chat or social logins, it focuses on delivering a smooth, responsive, and privacy-respecting user experience, prioritizing in-house functionalities to ensure platform independence and data security.

Table 5-1 External API

Component	Technology / Tool	Purpose / Description
Backend Framework	Laravel (PHP)	Handles server-side logic, routing, database interactions, and overall application structure.
AI-Based Try-On System	Computer Vision (OpenCV)	Performs facial landmark detection and virtual eyewear fitting using real-time webcam or uploaded images.
Frontend UI	Blade Templates + CSS	Provides a responsive and user-friendly interface for users to browse products and try on eyewear virtually.
Rich Text Editor	TinyMCE (or similar)	Allows admins to create and edit FAQs, product descriptions, and announcements easily within the admin panel.
Database	MySQL	Stores user data, product catalog, try-on session data, and admin settings.
Authentication	Laravel Authentication	Manages user registration, login, and session handling without third-party OAuth integrations.
Security	Laravel Middleware + CSRF	Ensures secure user sessions, protects against cross-site request forgery and common vulnerabilities.
Try-On Processing	Custom Laravel Services	Handles the facial analysis and overlays virtual frames on user images or live video feeds.
Admin Panel	Laravel + Blade	Provides management tools for products, users, and FAQs, controlled via secure admin routes.

5.3 User Interface

The Optical Facial Fit (OFF) user interface (UI) is designed with simplicity and user-friendliness in mind, featuring a clean, intuitive layout that makes it easy for users to navigate engagement.

5.3.1 Home Screen

This image illustrates the homepage of Optical Facial Fit (OFF), the main gateway for users looking to explore and virtually try on eyewear. The layout is designed to be welcoming and user-friendly, instantly showcasing the platform's key feature AI-powered virtual try-on technology. Prominent options for login and signup encourage user engagement, while intuitive navigation directs visitors to browse products, access their cart or favorites, and learn more about the services offered.

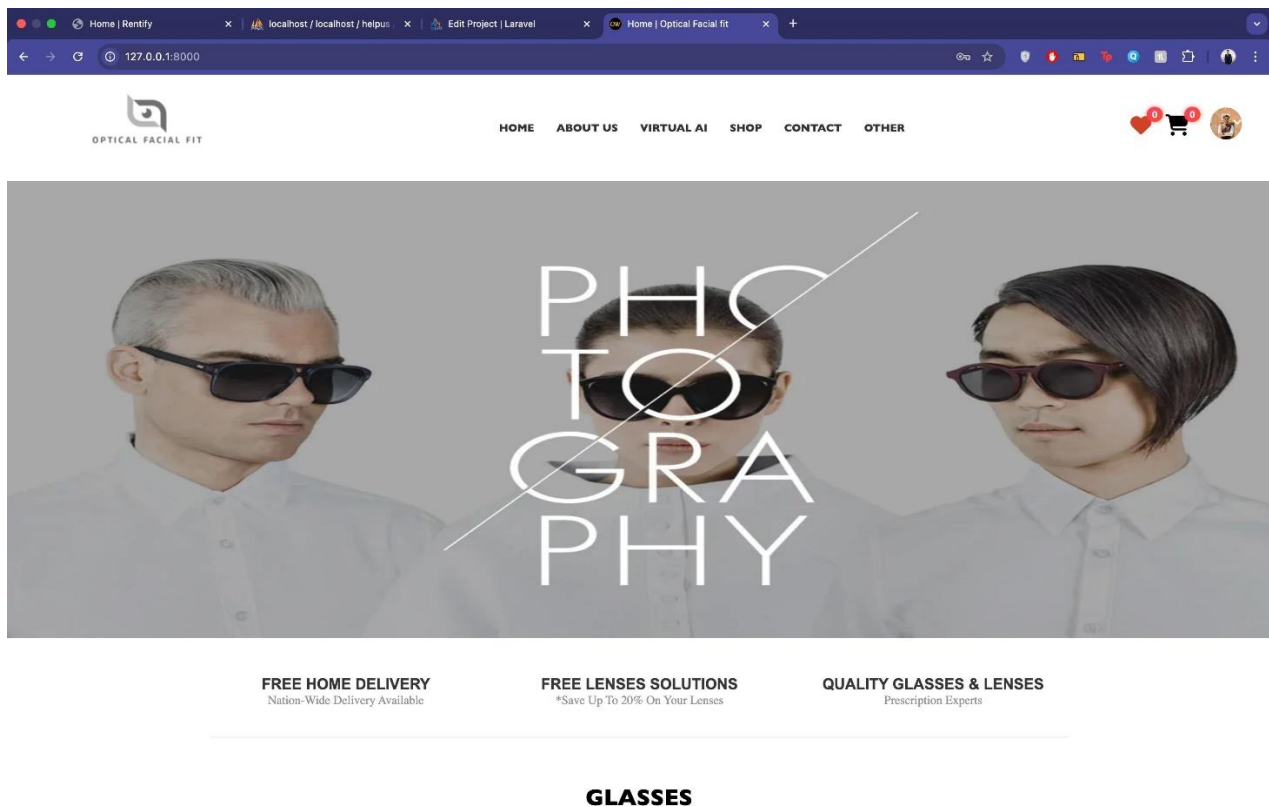


Figure 5.3-1 home page

5.3.2 Product catalog

Displays a browsable and searchable list of eyewear products.

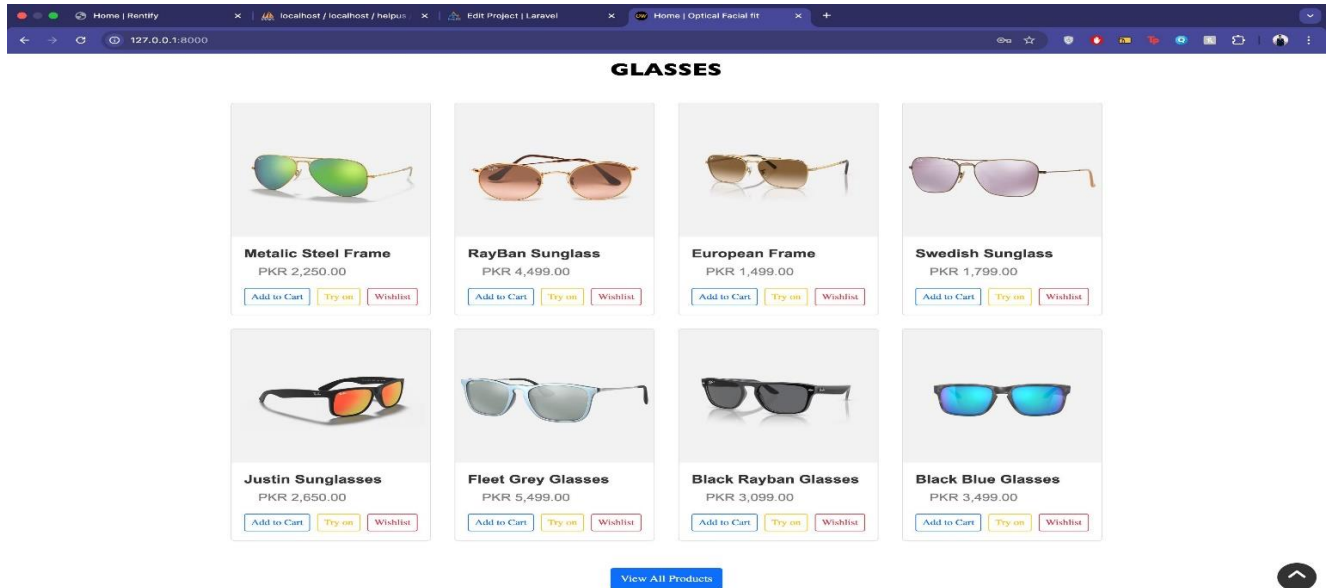


Figure 5.3-2 product catalog

5.3.3 AI try-on feature

Enables users to try on glasses virtually using facial image input and computer vision.

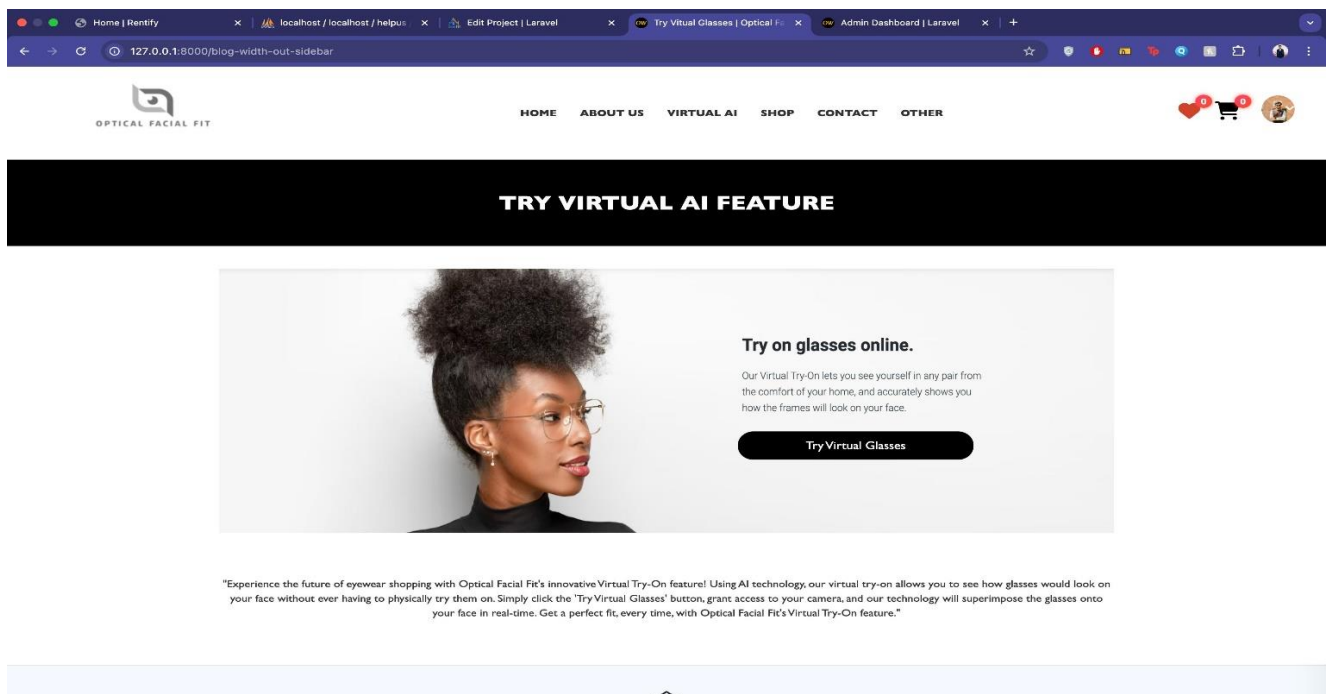


Figure 5.3-3 AI try-on feature

5.3.4 Customer Reviews

Enables users to read and submit feedback on products, helping others make informed purchasing decisions.

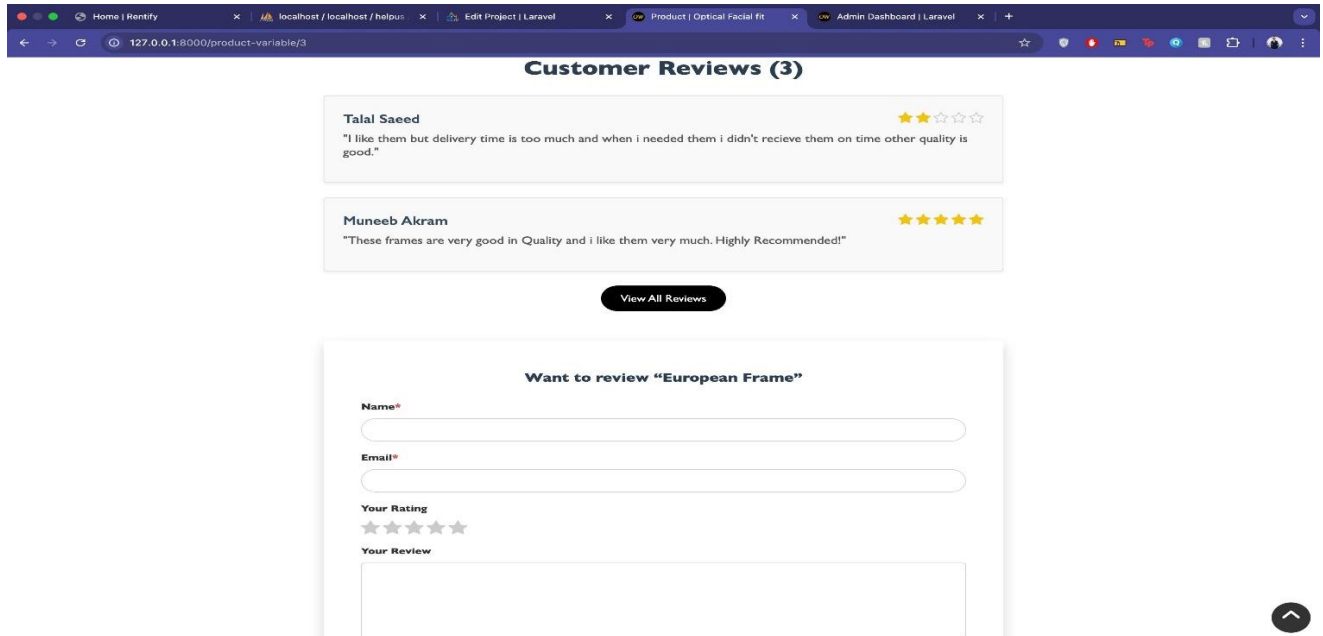


Figure 5.3-4 customer reviews

5.3.5 Product Detail Page

Displays detailed information about a selected frame including size, color options, price, material, and user reviews.

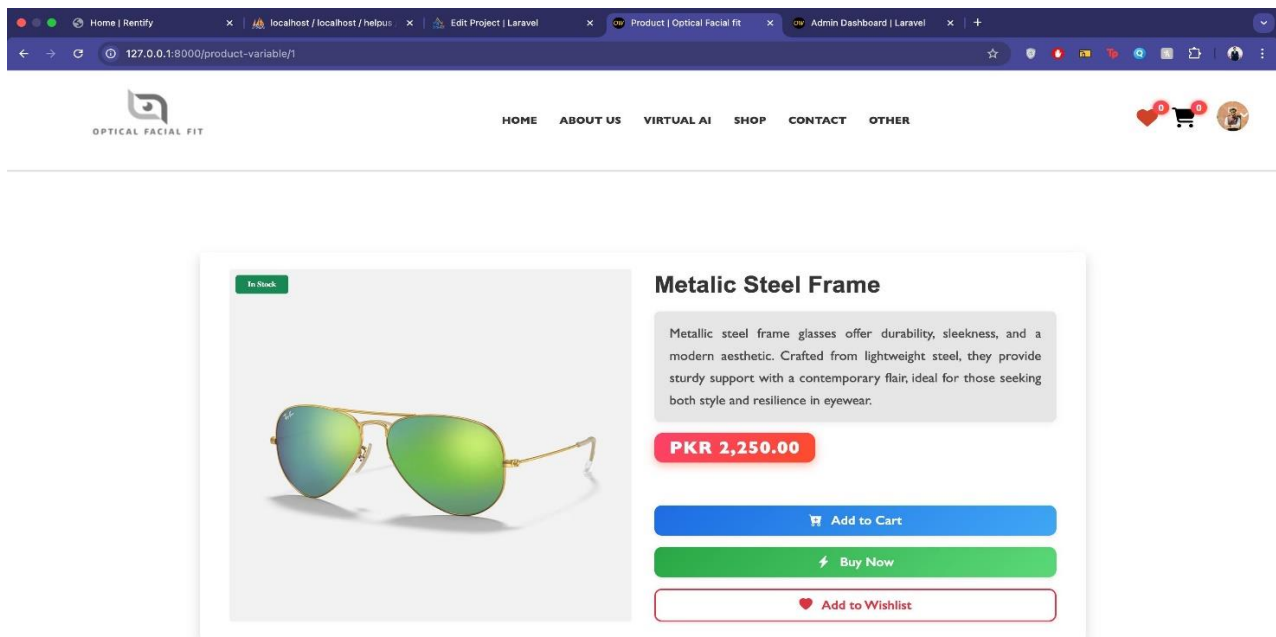


Figure 5.3-5 product card

5.3.6 Shopping cart

Allows users to view, manage, and proceed with selected eyewear items before checkout.

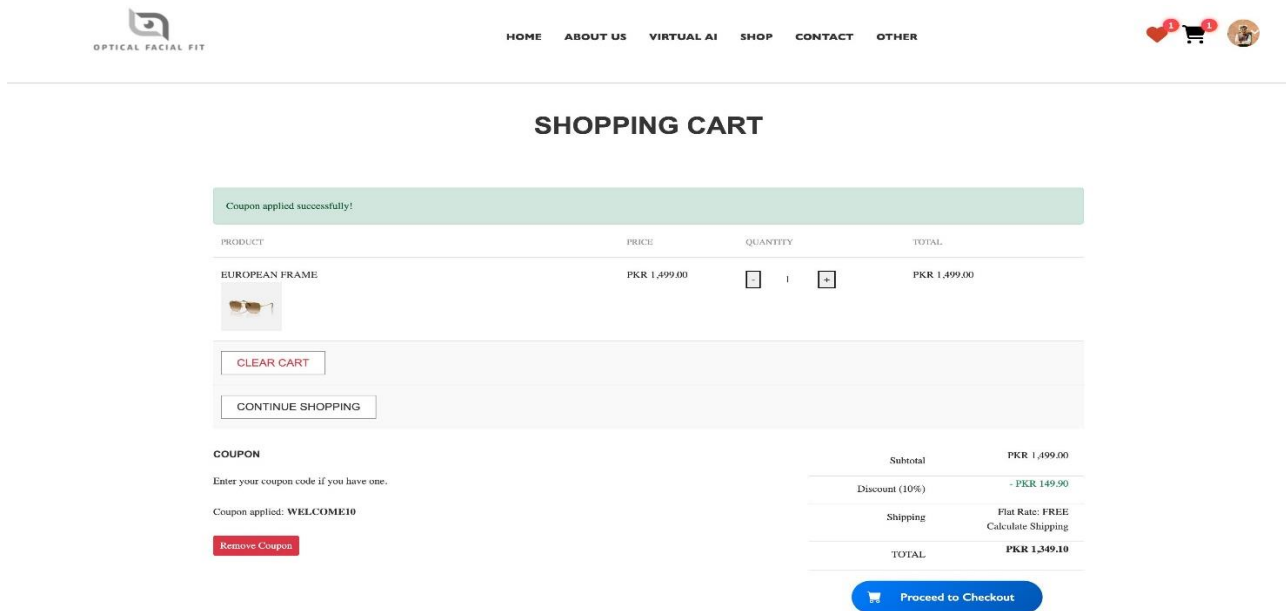


Figure 5.3-6 shopping cart

5.3.7 Wishlist Interface

Enables users to save their favorite eyewear products for future viewing or purchase.

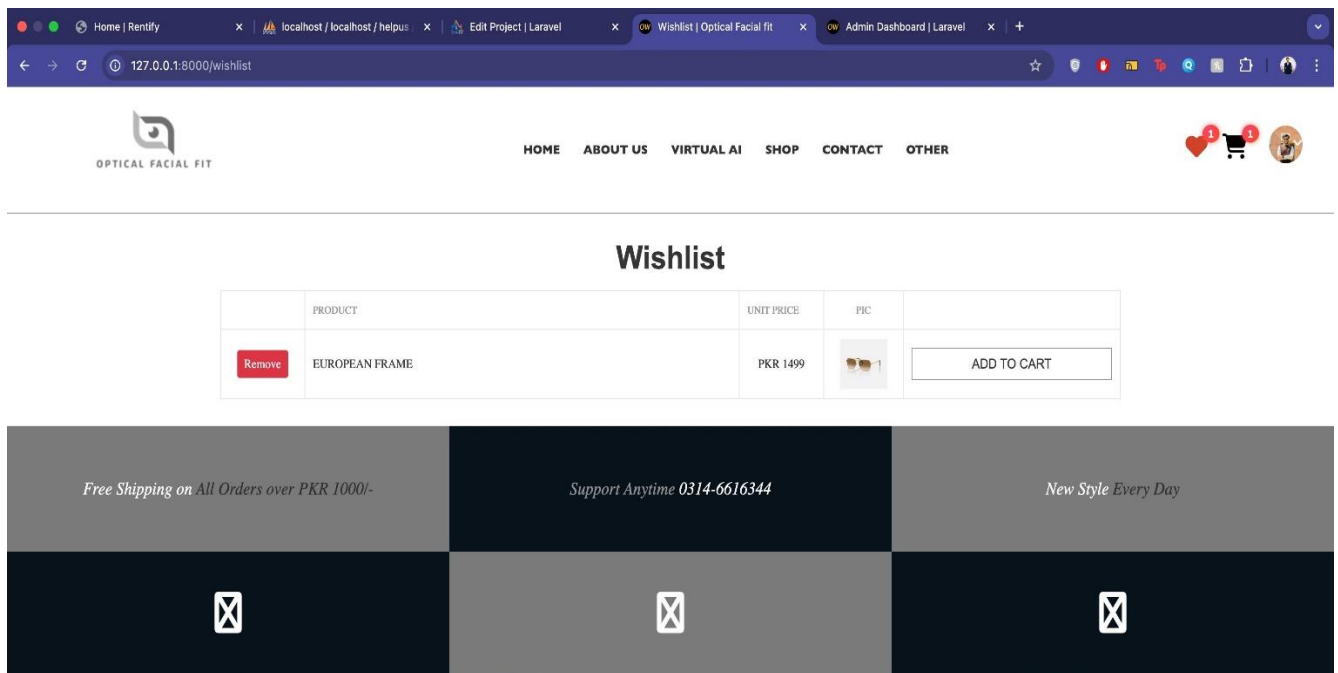


Figure 5.3-7 wishlist interface

5.3.8 Order Page Interface

Displays a summary of selected items, shipping details, and total cost, guiding users to finalize their purchase through checkout and payment.

YOUR ORDER

BILLING DETAILS
First Name * Last Name *
Email * Phone *
Address *
City * Postal Code * Country *

PRODUCT	TOTAL
EUROPEAN FRAME x 1	PKR 1,499
CART SUBTOTAL	PKR 1,499
DISCOUNT (10%)	- PKR 149.90
SHIPPING	Flat Rate: FREE
ORDER TOTAL	PKR 1,349.10

Choose a Payment Method

Figure 5.3-8 Order page interface

Your order has been successfully placed!

Thank you for your purchase. We will process your order and if your payment method is online kindly check "My Order Status" for payment status.

Order Summary
Order Number: #ORD-1747347001
Date: May 15, 2025
Total Amount: PKR 1,499

[Back to Home](#)

Figure 5.3-9 successful payment interface

5.3.9 Admin Panel

A secure backend dashboard accessible only to admins for managing product listings, user activity logs, customer queries, and overall system configurations.

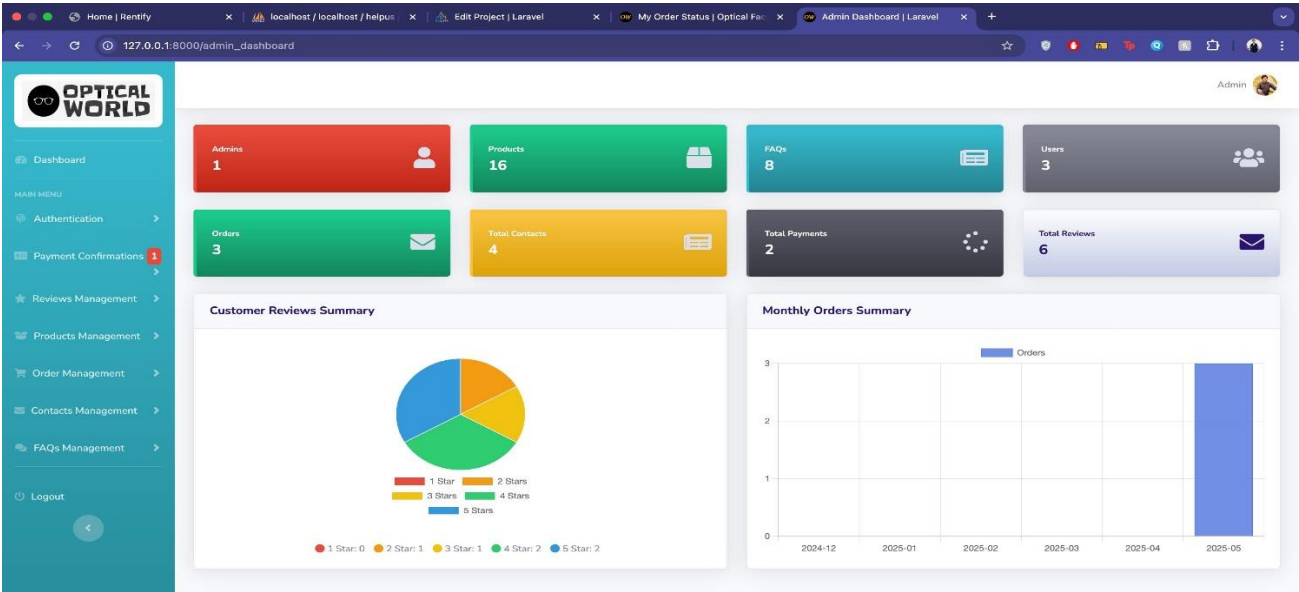


Figure 5.3-10 Admin Panel

5.3.10 Order Status Table

Provides customers with real-time updates on their order progress, including statuses like "Pending," "Processing," "Shipped," and "Delivered" for easy tracking.

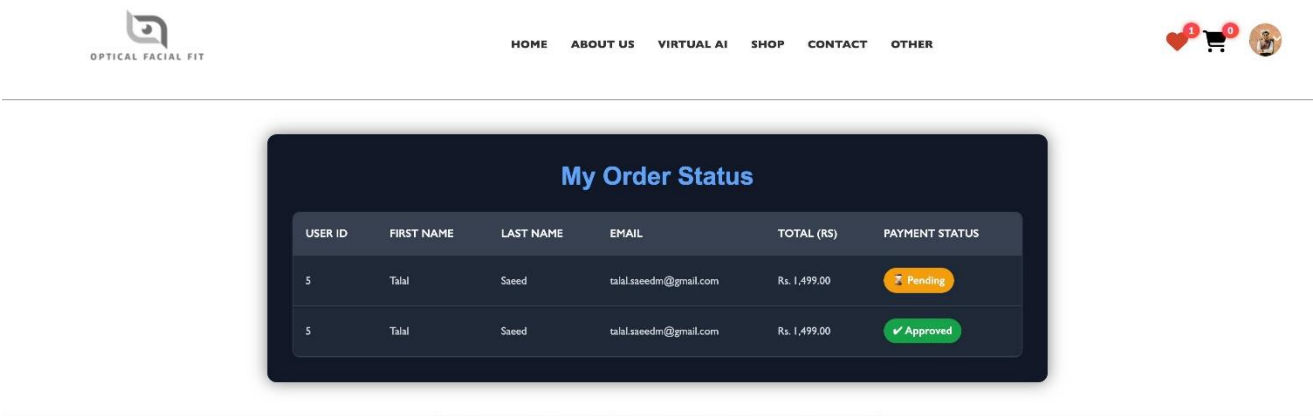


Figure 5.3-11 order status table

5.3.11 Admin Payment Approval Interface

Enables the admin to review transaction details and approve or reject payments to confirm and process customer orders.

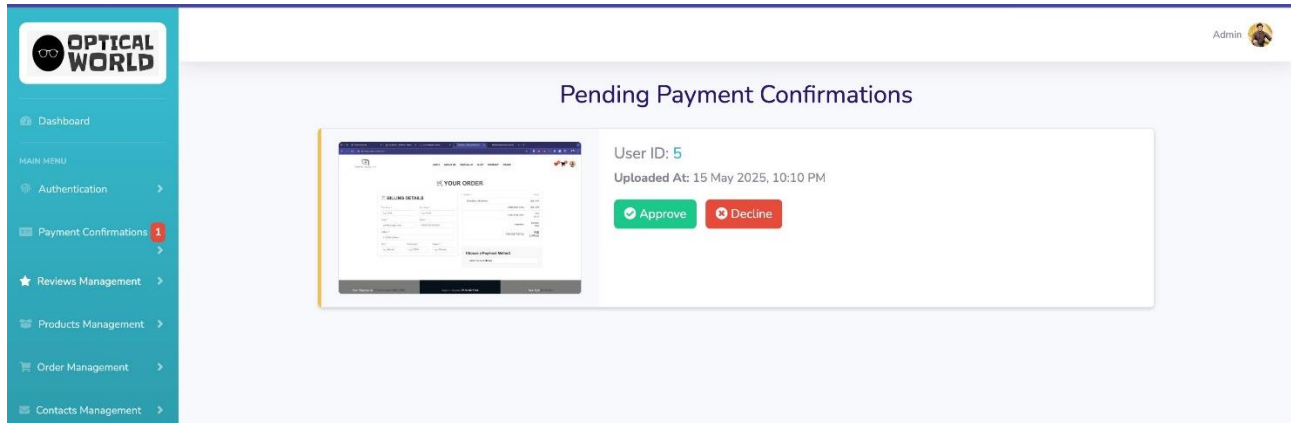


Figure 5.3-12 admin payment approval interface

5.3.12 Admin Review management

Allows the admin to view, moderate, approve, or delete customer reviews to maintain quality and relevance across the platform.

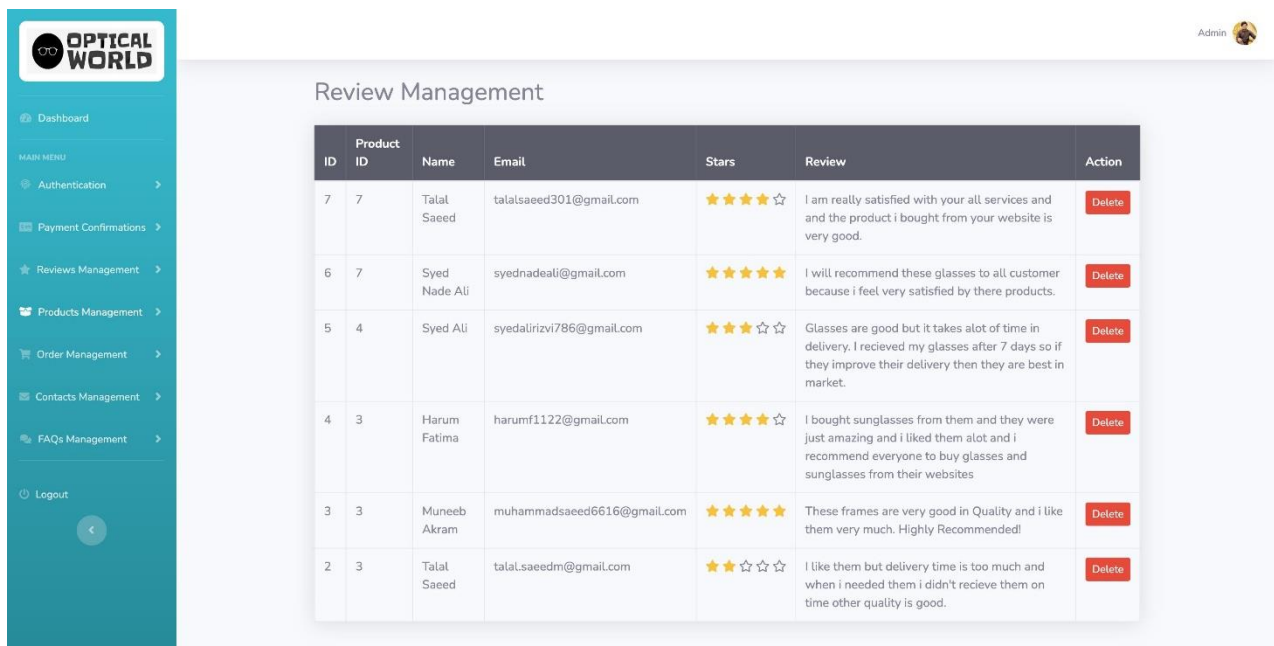


Figure 5.3-13 Admin review management

5.3.13 Add product page

Enables the admin to upload new eyewear products by entering details such as frame name, price, brand, style, description, and uploading corresponding images for display in the product catalog.

OPTICAL WORLD

Admin

Add Product

Title

Description

Price

Quantity

Category

Type

Product Image

Choose file No file chosen

Save

Figure 5.3-14 add product page

5.3.14 Placed Orders Page

Displays a list of all customer orders with relevant details such as customer name, order ID, products ordered, payment status, and order date, allowing the admin to track, approve, or update the order processing status.

OPTICAL WORLD

Admin

Placed Orders

Export to Excel Export to PDF

#	Name	Email	Phone	Address	Total (PKR)	Date
1	Talal Saeed	talaLsaeedm@gmail.com	03319696344	140-E Farid Town Sahiwal, Sahiwal, 57000, Pakistan	Rs 1,499.00	15 May 2025, 10:10 PM
2	Talal Saeed	talaLsaeedm@gmail.com	03319696344	140-E Farid Town Sahiwal, Sahiwal, 57000, Pakistan	Rs 1,499.00	15 May 2025, 06:08 PM
3	Talal Saeed	talaLsaeedm@gmail.com	03319696344	140-E Farid Town Sahiwal, Sahiwal, 57000, Pakistan	Rs 1,499.00	15 May 2025, 08:17 AM

Figure 5.3-15 placed order page

5.3.15 All Products Page

Provides a complete overview of all listed eyewear products, including details like product name, brand, category, price, stock status, and options to edit, delete, or update listings for efficient inventory management.

The screenshot displays the 'All Products' page of the Optical World dashboard. The page features a sidebar on the left with navigation links: Dashboard, Authentication, Payment Confirmations, Reviews Management, Products Management (active), Order Management, Contacts Management, and FAQs Management. The main content area shows a table of products with columns: Title, Price, Image, Category, Type, and Actions. A 'Delete' button is present in the Actions column for each product. The table lists 10 products, all categorized as 'Eye Glasses' and 'physical' type. The 'Add Product' button is located at the top left of the table.

Title	Price	Image	Category	Type	Actions
Scuderia Glasses	4899		Eye Glasses	Physical	Delete
Swedish Sunglass	1799		Eye Glasses	physical	Delete
Justin Sunglasses	2650		Eye Glasses	physical	Delete
Fleet Grey Glasses	5499		Eye Glasses	physical	Delete
Black Rayban Glasses	3099		Eye Glasses	physical	Delete
Black Blue Glasses	3499		Eye Glasses	physical	Delete
Blaze Cat Glasses	2799		Eye Glasses	physical	Delete
Blaze Shooter Glasses	4099		Eye Glasses	physical	Delete

Figure 5.3-16 All products page

Chapter 6

TESTING AND EVALUATION

6 Testing and Evaluation

Testing is a vital component in the development lifecycle of the Optical Facial Fit (OFF) platform, ensuring that the system functions reliably, securely, and efficiently across all user interactions. Given the platform's AI-based try-on feature, real-time facial alignment algorithms, and integrated e-commerce functionalities, a rigorous testing process is essential to deliver a seamless and error-free user experience. Testing is conducted at various stages of development from unit tests during feature implementation to comprehensive integration and user acceptance testing before deployment.

The primary goal of testing in OFF is to validate the accuracy of the virtual try-on experience, verify product recommendation logic, and ensure the stability of critical modules such as user authentication, cart and checkout workflows, and admin management tools. Special attention is also given to testing the AI-driven facial landmark detection and rendering process to maintain alignment precision across different face shapes, lighting conditions, and devices.

Following best practices in software engineering, testing in OFF is continuous and iterative. Each module undergoes:

- **Unit Testing** to validate individual functionalities such as login, product filtering, or AI recommendations.
- **Integration Testing** to verify that modules such as virtual try-on, cart system, and payment gateways interact correctly.
- **User Acceptance Testing (UAT)** to ensure that real users can navigate the system effectively and receive the intended experience.

Security testing is also performed to guard against threats such as unauthorized access, image manipulation, or data breaches. Additionally, performance testing is carried out to ensure that the platform handles high traffic during peak shopping periods without delays or failures.

By implementing a structured and layered testing strategy, the OFF platform ensures that any critical bugs, UI inconsistencies, or performance issues are identified early and addressed promptly. This approach results in a robust, intuitive, and high-performing platform that instills user confidence and trust in the virtual eyewear shopping experience.

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6.1 Manual Testing

Manual Testing is a crucial aspect of the testing process for the Optical Facial Fit (OFF) platform. It involves testers manually navigating the system to simulate real-world user interactions and validate

that all features perform as expected. During this process, testers explore key functionalities such as user login and registration, the AI-powered virtual try-on experience, product browsing, cart management, and the checkout flow. Special attention is given to ensuring that eyewear aligns accurately with the user's facial structure across different devices and lighting conditions.

6.1.1 System Testing

System Testing for Optical Facial Fit (OFF) is conducted after the development phase to ensure that all platform components work together as a unified, seamless system. This critical stage involves evaluating the integration of core features, including the AI-powered virtual try-on module, user authentication, product catalog navigation, cart and favorites management, payment processing, and the admin dashboard functionalities.

The objective of this phase is to confirm that OFF meets all specified requirements in terms of performance, security, visual accuracy, and user experience. System testing focuses on identifying potential issues that may arise from the interaction between modules—for example, misalignment of eyewear during virtual try-on, synchronization failures between cart and product inventory, or errors in the checkout or payment workflow.

By validating the complete end-to-end functionality in a real-world scenario, system testing ensures that the OFF platform delivers a stable, secure, and visually accurate environment. This guarantees users a reliable experience when exploring, trying on, and purchasing eyewear virtually before the platform is released for public use.

6.1.2 Unit Testing

Unit Testing in the Optical Facial Fit (OFF) platform focuses on verifying the functionality of individual components in isolation to ensure they perform as intended. This includes testing discrete units such as the user login and registration module, virtual eyewear rendering logic, product addition to cart/favorites, and admin panel controls.

For instance, unit tests are applied to functions that handle the AI-based frame fitting algorithm, image upload validation, and real-time preview adjustments. These tests check whether each component correctly handles inputs (such as face image or selected eyewear), processes data accurately (e.g., alignment and scaling of frames), and returns the expected outputs without reliance on other modules.

Developers implement unit tests for all critical backend services and frontend interactions to ensure robustness before integration. This proactive testing approach helps identify and resolve issues early

in the development cycle and smoother integration of features across the **OFF** platform.

6.2 Unit Testing

6.2.1 Unit Testing 1: Login as Admin

Testing Objective: Ensure the admin login form functions correctly.

Table 6-1: Admin Login

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
UT-001	Admin Login by entering username and password	Username: admin@offplatform.com Password: adminSecure@2025	Admin successfully logs into the system	Pass

6.2.2 Unit Testing 2: Login as a User

Testing Objective: Ensure the user login and registration system works as expected.

Table 6.2: User Login

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
UT-002	User Login by entering username and password	Username: user1@off.com Password: userPass@123	User successfully logs into the system	Pass

6.2.3 Unit Testing 3: Logout as Admin

Testing Objective: Ensure admin can log out successfully.

Table 6.3: Admin Logout

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
UT-003	Click the logout button	Logout	Admin successfully exits the system	Pass

6.2.4 Unit Testing 4: Logout as User

Testing Objective: Ensure user can log out successfully.

Table 6.4: User Logout

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
UT-004	Click the logout button	Logout	User successfully exits the system	Pass

6.3 Functional Testing

6.3.1 Functional Testing 1: Login with Different Roles

Objective: Ensure role-based access and navigation load correctly.

Table 6.5: Login as Different Roles

No.	Test Case	Attribute & Value	Expected Result	Result
1	Login as Admin	Username: admin@offplatform.com Password: adminSecure@2025	Admin dashboard loads with admin navigation	Pass
2	Login as Customer	Username: customer01@off.com Password: tryon2025	Customer homepage loads with try-on and product options	Pass
3	Login as Inventory Manager	Username: manager@off.com Password: stockAdmin2025	Product management dashboard loads with proper controls	Pass

6.4 Integration Testing

Objective: Ensure modules such as login, try-on, product browsing, and cart functionalities work cohesively.

Table 6.6: Integration Testing

No.	Test Case	Attribute & Value	Expected Result	Result
1	Login and access virtual try-on	Username: user1@off.com Password: userPass@123	User logs in and is redirected to try-on module	Pass
2	Add eyewear to favorites	Action: Click "Add to Favorites" on a selected frame	Item appears in the user's favorites list	Pass
3	Add eyewear to cart	Action: Click "Add to Cart" on a selected frame	Item successfully appears in the cart	Pass
4	Submit contact form	Name: John Doe Email: john@off.com Message: "Issue with virtual try-on"	Message is stored in admin contact log	Pass
5	Process checkout	Action: Click "Proceed to Payment" and confirm	User is redirected to payment gateway	Pass

Chapter 7

CONCLUSION AND FUTURE WORK

7 Conclusion

The Optical Facial Fit (OFF) platform offers an advanced and interactive solution for enhancing the online eyewear shopping experience through AI-driven technology. Designed and developed using PHP Laravel, the system integrates essential e-commerce functionalities, including user authentication, product management, contact forms, cart/favorites features, FAQs, and an admin panel for system oversight.

At the core of the platform lies a computer vision-based virtual try-on feature, which enables users to realistically preview how different eyeglass frames will appear on their faces. This implementation utilizes image processing algorithms to detect facial landmarks and accurately overlay eyewear models, providing users with a personalized and engaging experience.

Unlike many commercial platforms that rely on external APIs or third-party services, OFF is developed as a self-contained Laravel application, ensuring complete control over both frontend and backend logic. The use of MySQL for structured data storage and Cloudinary for optimized media management supports smooth performance and scalability.

The simple and responsive interface makes navigation effortless for users while maintaining administrative flexibility for managing products, monitoring logs, and handling customer interactions. By combining traditional e-commerce components with emerging computer vision techniques, Optical Facial Fit demonstrates a practical and innovative approach to solving real-world challenges in online retail.

The project effectively bridges the gap between digital convenience and physical product interaction, offering a reliable, accessible, and user-centric platform that caters to both customers and eyewear vendors.

7.1 Future Work

While the current version of Optical Facial Fit (OFF) successfully delivers a seamless eyewear try-on and shopping experience, several opportunities exist to further elevate the platform's functionality, scalability, and user engagement. Future work can focus on the following areas:

7.1.1 Enhanced AI-Based Try-On Accuracy

- Integrating more advanced facial landmark detection models to improve fitting precision and support a wider variety of face shapes and lighting conditions.

- Incorporating 3D try-on technology using deep learning and depth estimation to simulate realistic frame fitting with head movements.

7.1.2 Frame Recommendation System

To enhance personalization and user satisfaction, OFF aims to develop a machine learning–based recommendation engine that intelligently suggests eyewear frames tailored to individual users. This system would analyze a combination of factors including detected face shape (e.g., oval, round, square), user demographics, previous try-on history, browsing patterns, and feedback data. Using algorithms such as collaborative filtering and content-based filtering, the engine can continuously learn from user interactions to provide increasingly accurate recommendations.

7.1.3 Mobile Application Integration

To expand platform accessibility and improve user convenience, OFF plans to develop a native mobile application for both Android and iOS devices. This mobile app will offer seamless access to the virtual try-on experience, allowing users to explore and try eyewear from anywhere, using their smartphone cameras. By leveraging native capabilities like real-time camera access, device-level notifications, and gesture-friendly UI components, the mobile app will deliver a smoother and more interactive experience compared to mobile web browsers. It will also support features such as saving favorites, in-app purchases, and personalized recommendations, ensuring users have full access to the platform’s core functionalities while on the go.

7.1.4 Augmented Reality (AR) Integration

To elevate user interaction and eliminate friction in the eyewear selection process, OFF aims to integrate real-time Augmented Reality (AR) support. This feature will allow users to virtually try on glasses using their mobile device cameras without needing to upload photos manually. Leveraging computer vision and facial landmark detection, the AR system will accurately overlay frames on the user’s face in real time, adapting to movements and facial angles. This hands-free, dynamic experience not only enhances convenience but also increases confidence in purchase decisions by providing a highly realistic preview.

7.1.5 Multi-Vendor Support

To broaden the reach and scalability of the Optical Facial Fit (OFF) platform, future development will focus on supporting multiple sellers or optical retailers. This enhancement will enable various optical

stores to register, list, and manage their own eyewear product catalogs within a shared marketplace environment. Each vendor will have access to a dedicated dashboard for handling inventory, pricing, orders, and customer interactions. This multi-vendor model not only diversifies the range of available eyewear options for users but also promotes competition and collaboration among sellers. By turning OFF into a centralized hub for optical products, the platform can cater to a wider customer base while supporting business growth for independent eyewear brands.

7.1.6 Advanced Admin Analytics

Future improvements will include upgrading the admin panel to incorporate advanced visual analytics, enabling administrators to gain deeper insights into platform performance and user engagement. This includes integrating dashboards that display real-time data on user behavior, such as most-tried frames, peak usage times, and conversion rates.

7.1.7 Accessibility and Localization

Ensure a more inclusive user experience, future development will focus on enhancing accessibility features across the Optical Facial Fit platform. This includes implementing screen reader compatibility, keyboard navigation, and a high-contrast mode for visually impaired users. Additionally, introducing multi-language support will broaden the platform's usability for a global audience, allowing users to navigate the interface and try on eyewear in their preferred language.

7.1.8 Integration with Payment Gateways

Expanding current payment support with integrations for popular gateways like Jazzcash, Easypaisa, or Card payment for smoother transactions.

Through continuous iteration and user-centered improvements, OFF can evolve into a robust platform that not only facilitates online eyewear purchases but also redefines how customers interact with fashion and technology.

Chapter 8

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